

Supplementary Material — The Nostalgic Undertow

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This document contains the complete proofs of Theorems 1–3 of the main text and the technical details of the numerical simulations of § 6. Notation and definitions are inherited from [1] without redefinition: $K = k(k-1)$ is the dimension of the logits of the OU parametrization (2.1) main, $\tau_E = 1/\lambda$ is the characteristic drift time, $\sigma_0^2 = \sigma^2/(2\lambda)$ is the stationary variance of the logit, $\epsilon := \sigma^2/\lambda$ is the OU-concentration parameter (of the latent itself), and $\epsilon_{\text{track}} := \mathcal{I}_{\text{rate}}\sigma^2/\lambda^2$ is the *tracking* control parameter (the concentration of the filter’s estimate of the latent, (2.2) main).

S1. Proof of Theorem 1 (the adiabatic identity $B = K/2$ and its transient realizability, OU)

Theorem 1 main states that in the adiabatic OU limit the cumulative excess loss grows as $L_{\text{excess}}(t) = At + (K/2)\ln(\lambda t) + C + o(1)$ with logarithmic coefficient equal to $K/2$, **on the transient concentration window** $t \lesssim \tau_{\text{conc}}$ (the duration of the prior-collapse burst, S1.3a–c); asymptotically, for $t \gg \tau_{\text{conc}}$, the steady-state gain g^* saturates the accumulation and the logarithm passes into a constant plateau $\sim (K/2)\ln(1/g^*)$ (S1.8a) — a structural transience, analyzed in S1.4–S1.5. The derivation is carried out in the linear-Gaussian (Kalman) approximation of the observation model and under the explicit assumptions (A1)–(A3) of the statement of Theorem 1 main; in particular, step (S1.4) rests on the heuristic reduction $n_{\text{eff}} \asymp \lambda s$ to class II (A3) *on the transient window*, while a rigorous transient non-stationary filter is left open (§ 7 main). Within this framework the coefficient $K/2$ does not depend on σ or on the curvature of the model. The derivation proceeds in six steps: (S1.1) reduction to a scalar tracking problem over independent coordinates; (S1.2) tracking error of the EWMA filter in an OU environment, the Riccati solution, the steady-state g^* and two time scales (τ_f^{forget} of forgetting and τ_{conc} of concentration); (S1.3) expansion of the instantaneous excess loss to a quadratic form in the Fisher metric; (S1.4) integration on the transient window, derivation of the coefficient $K/2$ and the transition of the logarithm into a plateau for $t \gg \tau_{\text{conc}}$; (S1.5) analysis of the finite-amplitude correction $g(\epsilon)$ and the structural unattainability of $\hat{B} \rightarrow K/2$ as a slope; (S1.6) optimality of the floor as a lower bound over all memory architectures (sufficiency of the Kalman posterior + MMSE, in the linear-Gaussian window A1).

S1.1 Tensorization over independent coordinates

By construction of the OU parametrization (2.1) main, the increments dW_{ij} are independent over (i, j) , the covariance is diagonal, and the process $\theta_v(t) = (\theta_{ij}(t))$ decomposes into $K = k(k-1)$ independent scalar OU processes

$$d\theta_{ij} = -\lambda(\theta_{ij} - \theta_{ij}^*) dt + \sigma dW_{ij}, \quad \mathbb{E}[\theta_{ij}(t)\theta_{ij}(s)] = \sigma_0^2 e^{-\lambda|t-s|}.$$

For the ideal Bayesian learner, the posterior over $\theta_v(t)$ factorizes across coordinates to leading order in the adiabatic limit (per-bit uniformity, [1] Remark 5, § S2);

consequently, both the instantaneous excess loss and its cumulative sum are additive:

$$L_{\text{excess}}(t) = \sum_{ij} L_{\text{excess}}^{(ij)}(t) \cdot (1 + o(1)), \quad (\text{S1.1})$$

where $L_{\text{excess}}^{(ij)}$ is the excess loss on a single coordinate. The additivity of mutual information and KL divergence over independent coordinates is a standard identity [2] ch. 2.5; unlike the χ^2 measure (multiplicative under independence), KL/MI are additive, which is precisely what yields the passage to the factor K in the final coefficient (the same logic as in [1] § S1.5). It suffices to prove the scalar result $L_{\text{excess}}^{(ij)}(t) = at + \frac{1}{2} \ln(\lambda t) + c + o(1)$ and to sum over the K coordinates.

Coordinate-free derivation of the coefficient $K/2$ (the load-bearing argument). The component-wise tensorization (S1.1) is convenient as an illustration, but the *value* of the logarithmic coefficient $K/2$ does not depend on it and does not rely on the diagonality of the Fisher matrix or on the independence of coordinates. The logarithmic term of the transient concentration (S1.4–S1.8 below) is, in matrix form, the trace of the product of the Fisher metric with the transient posterior covariance: when the posterior concentrates with effective number of observations n_{eff} , the transient covariance in θ_v is $\Sigma_{\text{tr}} = (\mathcal{I} n_{\text{eff}})^{-1}$ (Bayesian concentration for a K -dimensional parameter), and the instantaneous logarithmic contribution to the excess loss is the quadratic form (S1.5) with trace

$$\frac{1}{2} \text{tr}(\mathcal{I} \cdot \Sigma_{\text{tr}}) = \frac{1}{2} \text{tr}(\mathcal{I} (\mathcal{I} n_{\text{eff}})^{-1}) = \frac{1}{2} \text{tr}(\text{Id}_K) / n_{\text{eff}} = \frac{K}{2 n_{\text{eff}}},$$

since $\mathcal{I} (\mathcal{I})^{-1} = \text{Id}_K$ for *any* SPD Fisher matrix $\mathcal{I}$ (diagonal or not), and $\text{tr}(\text{Id}_K) = K$. The Fisher information cancels *identically* — inside the trace, without appeal to a coordinate-wise decomposition — so the coefficient $K/2$ holds even when the softmax couples the logits of a single row (Fisher non-diagonal): it depends only on the dimension $K = \dim \theta_v$, and not on the structure of $\mathcal{I}$ or on per-bit uniformity. Substituting $n_{\text{eff}} \asymp \lambda s$ (S1.4) and integrating $\int (K/2) ds/s$ yield (S1.10) directly. Per-bit-uniformity remains needed only for the *floor* C2 (§ S2, additivity of the contribution of fresh bits), but *not* for the value of $K/2$. The coordinate-wise derivation (S1.1)–(S1.10) below is a particular (diagonal) case of this trace, given for clarity.

S1.2 Tracking error of the EWMA filter and the Riccati equation

Consider a scalar coordinate $\theta_v(t)$ (we drop the index ij) with hidden OU dynamics $d\theta_v = -\lambda \theta_v dt + \sigma dW$ (anchor $\theta_v^* = 0$ without loss of generality). Observations are discrete (one measurement per step $\Delta t = 1$, as in the simulations). In the linear-Gaussian approximation around the current estimate, an observation is equivalent to a noisy measurement $dy_s = \theta_v(s) ds + \mathcal{I}_{\text{rate}}^{-1/2} dV_s$, where V_s is a standard Wiener process and $\mathcal{I}_{\text{rate}}$ is the *rate of accumulation of Fisher information* (Fisher rate) of dimension [Fisher]·time⁻¹: the amount of Fisher information about θ_v arriving per unit time. Under the discretization $\Delta t = 1$ with one observation per step, $\mathcal{I}_{\text{rate}} = \mathcal{I}$ numerically

(information per step equals information per unit time), and below we write $\mathcal{I}$ in the sense of $\mathcal{I}_{\text{rate}}$; in the general case the correspondence is $\mathcal{I}_{\text{rate}} = \mathcal{I}/\Delta t$.

The optimal Bayesian filter for a linear-Gaussian system with hidden OU state is the stationary Kalman–Bucy filter [3, 4]. The stationary posterior variance (tracking error) $\Sigma_\infty := \lim_{t \rightarrow \infty} \mathbb{E}[(\hat{\theta}_v(t) - \theta_v(t))^2]$ is the positive root of the continuous-time algebraic Riccati equation (with Fisher-rate $\mathcal{I}_{\text{rate}}$; the dimensions are consistent: $[\lambda\Sigma] = [\sigma^2] = [\mathcal{I}_{\text{rate}}\Sigma^2]$ — variance·time^{−1})

$$-2\lambda\Sigma_\infty + \sigma^2 - \mathcal{I}_{\text{rate}}\Sigma_\infty^2 = 0 \implies \Sigma_\infty = \frac{-2\lambda + \sqrt{4\lambda^2 + 4\mathcal{I}_{\text{rate}}\sigma^2}}{2\mathcal{I}_{\text{rate}}} = \frac{\lambda}{\mathcal{I}_{\text{rate}}} \left(\sqrt{1 + \frac{\mathcal{I}_{\text{rate}}\sigma^2}{\lambda^2}} - 1 \right). \quad (\text{S1.2})$$

The factor 2 in the term $-2\lambda\Sigma_\infty$ is standard for OU (the generator $-\lambda$ acts on the variance with doubling); the controlling dimensionless parameter of *tracking* convergence is exactly $\epsilon_{\text{track}} := \mathcal{I}_{\text{rate}}\sigma^2/\lambda^2$ (definition (2.2) main), that is, the ratio of the Fisher-rate to the drift rate, weighted by the noise amplitude. We emphasize the distinction from $\epsilon = \sigma^2/\lambda$: ϵ is the concentration of the OU latent itself, while $\epsilon_{\text{track}} = \mathcal{I}_{\text{rate}}\epsilon/\lambda$ is the concentration of the filter’s *estimate* of the latent; it is ϵ_{track} , not ϵ , that controls the expansion below. In the adiabatic limit $\epsilon_{\text{track}} \ll 1$, expanding the square root gives

$$\Sigma_\infty = \frac{\lambda}{\mathcal{I}} \cdot \frac{\mathcal{I}\sigma^2}{2\lambda^2} \left(1 + O\left(\frac{\mathcal{I}\sigma^2}{\lambda^2}\right) \right) = \frac{\sigma^2}{2\lambda} (1 + O(\epsilon_{\text{track}})) = \sigma_0^2 (1 + O(\epsilon_{\text{track}})). \quad (\text{S1.3})$$

This is the *stationary* residual lag of the filter: $\Sigma_\infty = \sigma_0^2(1 + O(\epsilon_{\text{track}})) \propto \sigma^2/\lambda \rightarrow 0$ in the OU-concentration limit. The equivalent *forgetting scale* of the filter $\tau_f^{\text{forget}} = 1/\sqrt{\lambda^2 + \mathcal{I}\sigma^2} \asymp 1/\lambda = \tau_E$ in the adiabatic limit — that is, the optimal filter forgets on the scale of the environment’s drift. (*Terminological refinement: below, two scales are distinguished. This one — the forgetting scale $\tau_f^{\text{forget}} \asymp \tau_E$ — sets the memory depth of the filter. A separate concentration scale $\tau_{\text{conc}} = n_{\text{eff}}^* = 1/g^*$ (the effective-observation floor, S1.3b) sets the scale of reaching the stationary g^* ; in the adiabatic limit they diverge as $\tau_{\text{conc}}/\tau_f^{\text{forget}} = \lambda/g^* = 2/\epsilon_{\text{track}} \rightarrow \infty$ (since $g^* = \frac{1}{2}\epsilon_{\text{track}}\lambda$, S1.3a).)*

Steady-state gain and the concentration floor. The stationary Kalman filter has a steady-state gain $g^* := \mathcal{I}_{\text{rate}}\Sigma_\infty/(1 + \mathcal{I}_{\text{rate}}\Sigma_\infty \Delta t)$, whose leading term in the adiabatic limit at $\Delta t = 1$ is

$$g^* = \mathcal{I}_{\text{rate}}\Sigma_\infty (1 + O(\epsilon_{\text{track}})) = \mathcal{I}_{\text{rate}} \cdot \frac{\sigma^2}{2\lambda} (1 + O(\epsilon_{\text{track}})) = \frac{1}{2} \epsilon_{\text{track}} \lambda (1 + O(\epsilon_{\text{track}})) \propto \sigma^2, \quad (\text{S1.3a})$$

that is, $g^* \rightarrow 0$ in the adiabatic limit, $g^* \propto \sigma^2$ (linear in the forcing variance at fixed $\lambda, \mathcal{I}_{\text{rate}}$). This g^* sets the *floor of the effective number of observations*: the stationary filter weights the past geometrically with factor $(1 - g^*)$ per step, so the effective number of accumulated informative observations saturates at

$$n_{\text{eff}}^* = \frac{1 - g^*}{g^*} \approx \frac{1}{g^*} \quad (\text{exactly } (1 - g^*)/g^*, \text{ leading term } 1/g^*), \quad (\text{S1.3b})$$

and does not grow without bound. The characteristic scale of reaching this floor is the *concentration scale* τ_{conc} , identified with the effective-observation floor (the number of steps over which n_{eff} reaches the plateau):

$$\tau_{\text{conc}} = n_{\text{eff}}^* = \frac{1 - g^*}{g^*} \approx \frac{1}{g^*} \propto 1/\sigma^2, \quad (\text{S1.3c})$$

the number of effective observations gathered by the time the floor is reached. Crucially: the floor $n_{\text{eff}}^* \asymp 1/g^* \propto 1/\sigma^2$ is a *different* scale from the forgetting scale $\tau_f^{\text{forget}} \asymp \tau_E = 1/\lambda$; in the adiabatic limit their ratio diverges: $\tau_{\text{conc}}/\tau_E = (1/g^*)/(1/\lambda) = \lambda/g^* = 2/\epsilon_{\text{track}} \rightarrow \infty$ (since $g^* = \frac{1}{2}\epsilon_{\text{track}}\lambda$, S1.3a). It is τ_{conc} , not τ_E , that bounds the window of logarithmic growth (S1.4), and its upper bound in adiabatic time is $\lambda\tau_{\text{conc}} = \lambda/g^* = 2/\epsilon_{\text{track}}$.

Origin of the upper bound of the window $\sim 1/\sigma^2$. It arises not from a “diffusive escape” of the latent: the OU restoring force keeps θ_v within the stationary spread $\sigma_0^2 = \sigma^2/2\lambda \ll 1$, reached already at $\tau_E = 1/\lambda \ll 1/\sigma^2$, so that the softmax linearization (A1) is valid at *all* t (its domain is determined by the *range* of the latent $\sigma_0 \ll 1$, not by the growing $\sigma^2 t$; the latter estimates the *free* random walk, not the bound OU). The upper bound is the scale of *tracking saturation*: the linear drift term $A \cdot t$ with $A = \frac{1}{2}\mathcal{I}_{\text{rate}}\Sigma_\infty \propto \sigma^2$ becomes comparable to the logarithm $(K/2)\ln(\lambda t)$ at $t \sim 1/\sigma^2$, after which the logarithm is statistically inseparable in the fit; equivalently, the steady-state $g^* \propto \sigma^2$ caps n_{eff} at the floor $n_{\text{eff}}^* \asymp 1/g^* \propto 1/\sigma^2$ (S1.3). Thus $1/\sigma^2$ is the scale of tracking saturation, not of linearization breakdown.

Why slope convergence is not achieved by the adiabatic limit (precise statement). The load-bearing no-go has already been established rigorously above: $\Sigma_\infty > 0$ (the Riccati root (S1.2), the MMSE optimum S1.6) caps n_{eff} at the floor $n_{\text{eff}}^* = 1/g^* < \infty$, so the concentration logarithm is *transient* — it grows only on the concentration window $t \lesssim \tau_{\text{conc}}$ and passes into a plateau for $t \gg \tau_{\text{conc}}$ (S1.4, S1.6c). Hence *any* wide asymptotic window $t \gg \tau_{\text{conc}}$ lies in the plateau and carries no log-slope — for *any* σ . It remains to dismiss the working hypothesis “as $\epsilon_{\text{track}} \rightarrow 0$ the logarithm will enter the asymptotic window as a slope”. The naive formulation of the dismissal — “ τ_{conc} and the ceiling $1/\sigma^2$ are both $\propto 1/\sigma^2$, hence they do not separate” — is *physically incomplete*: the mere coincidence of the powers of σ does not entail inseparability. The correct argument goes through the σ -invariance of the *asymptotic (plateau) fit window*. This window is $[t_{\min}, t_{\max}]$ with lower bound $t_{\min} = 3/g^* = 3n_{\text{eff}}^*$ (several floor scales n_{eff}^* , (S1.3b) — the operational scale at which the filter reaches the plateau: on it Σ_{tr} has already merged with Σ_∞ , the logarithm is saturated, S1.8a) and upper bound $t_{\max} = c_0/\sigma^2$ (the adiabatic ceiling); the log-bearing transient lies, by contrast, *below* $t_{\min}$ — the filter’s concentration completes on the scale $\sim \tau_E \ll t_{\min}$ (§ S4.1; the A3 heuristic $n_{\text{eff}} = \lambda s$ only *overestimates* the concentration time, the operational scale being the floor $\tau_{\text{conc}} = 1/g^*$,

§ 7 item iv), so the fit window does not intersect it (this is precisely how the scan of exp. (C) § S4.1 is set up). Its *logarithmic width* (the ratio of the bounds)

$$W = \frac{t_{\max}}{t_{\min}} = \frac{c_0/\sigma^2}{3/g^*} = \frac{c_0}{3} \frac{g^*}{\sigma^2} = \frac{c_0}{6} \frac{\mathcal{I}_{\text{rate}}}{\lambda} = \frac{c_0}{6} \frac{1}{\lambda r} \asymp \frac{1}{\lambda r}$$

(both bounds — $t_{\min} = 3/g^* = 3n_{\text{eff}}^*$ and $t_{\max} = c_0/\sigma^2$ — are numbers of *physical* steps, both $\propto 1/\sigma^2$ (the floor $1/g^* \propto 1/\sigma^2$); we substituted $g^* = \frac{1}{2}\mathcal{I}_{\text{rate}}\sigma^2/\lambda = \sigma^2/(2\lambda r)$ from (S1.3a), so that $g^*/\sigma^2 = 1/(2\lambda r)$ and σ^2 *cancels*; $r = 1/\mathcal{I}_{\text{rate}}$ is the observation noise). The decisive point: **the log-width of the asymptotic window** $W \propto 1/(\lambda r)$ **is σ -invariant** — σ has dropped out of it. Numerical caveat: the symbol $\asymp$ hides an $O(1)$ constant of the window shape $c_0/6 = 1/60$ (ceiling $c_0 = 0.1$, lower bound $3n_{\text{eff}}^* = 3/g^*$, factor $\frac{1}{2}$ in g^*), so that at $\lambda = r = 10^{-2}$ the actual value is $W = 1/(60\lambda r) \approx 167$, and *not* the bare $1/(\lambda r) = 10^4$; the numerical scan § S4.1 gives $W \in [156, 166]$, and what is confirmed is precisely the σ -invariance and the *scaling* $\propto 1/(\lambda r)$, not the absolute value. Therefore the adiabatic limit $\sigma \rightarrow 0$ ($\epsilon_{\text{track}} \rightarrow 0$) *does not shift the fit window out of the plateau toward the transient logarithm*: both the lower bound $t_{\min} = 3/g^* \propto 1/\sigma^2$ and the ceiling $\propto 1/\sigma^2$ stretch *in the same proportion*, the fit window keeps a fixed shape and stays entirely in the plateau. This *refutes* the working hypothesis (“push $\epsilon_{\text{track}} \rightarrow 0$ — the slope will be restored”): decreasing σ does not bring the fit window closer to the log-bearing transient. One can widen W only by increasing the Fisher-rate per correlation interval of the environment, $\mathcal{I}_{\text{rate}}/\lambda = 1/(\lambda r) \rightarrow \infty$ (infinitely precise observation $r \rightarrow 0$, or $\lambda \rightarrow 0$ faster than $\mathcal{I}_{\text{rate}}$). But this *takes one out of the slow-drift (adiabatic) regime*: $\mathcal{I}_{\text{rate}}/\lambda \rightarrow \infty$ means that $\epsilon_{\text{track}} = \mathcal{I}_{\text{rate}}\sigma^2/\lambda^2 = (\mathcal{I}_{\text{rate}}/\lambda)(\sigma^2/\lambda)$ ceases to be small at fixed latent concentration, violating the slow-driving condition $\lambda\tau_{\text{relax}} \ll 1$ and the quasi-stationarity of θ_v on which (A1) and the entire derivation rest. Conclusion: the separation of the log-bearing transient ($t \lesssim \tau_{\text{conc}}$) and the asymptotic fit window ($t \gg \tau_{\text{conc}}$) is a *consequence of the adiabatic restriction* (fixed Fisher-rate per correlation interval), and not of an arithmetic coincidence of the powers of σ ; in any regime where (A1) and quasi-stationarity hold, $W \asymp 1/(\lambda r) = O(1)$ with respect to σ , and the fit window does not draw in the log-bearing transient. This is the key to the structural unattainability of S1.5.

Beyond the stationary part, the filter has a *transient* component: starting from prior uncertainty, the posterior variance $\Sigma(t)$ decays toward Σ_∞ . For slow drift, the relevant transient variance is controlled by the accumulation of *informative* observations. **In the transient concentration regime** ($s \lesssim \tau_{\text{conc}}$, before reaching the steady-state g^*) observations separated by more than τ_E decorrelate (the covariance of θ_v decays as $e^{-\lambda|t-s|}$), and the effective number of independent observations about the current $\theta_v(t)$ grows as

$$n_{\text{eff}}(s) \asymp \frac{s}{\tau_E} = \lambda s \quad (\text{transient, } s \lesssim \tau_{\text{conc}}). \quad (\text{S1.4})$$

This is a *transient* (burst) regime: the growth (S1.4) does not continue up to $t \rightarrow \infty$ — at $s \gtrsim \tau_{\text{conc}}$ the steady-state g^* (S1.3a) caps the accumulation, and n_{eff} saturates at

the floor $n_{\text{eff}}^* = (1 - g^*)/g^* \approx 1/g^*$ (S1.3b). The linear growth λs and the saturation at $1/g^*$ coincide at $s \asymp \tau_{\text{conc}}$, which fixes the moment of transition. *Refinement of the concentration scale (to avoid an apparent double definition).* By τ_{conc} we mean throughout the *operational scale at which n_{eff} reaches the floor* — $\tau_{\text{conc}} = n_{\text{eff}}^* = 1/g^*$ steps (S1.3c): it is precisely this scale that anchors the lower bound of the fit window $t_{\text{min}} = 3/g^*$ (§ S4.1) and is consistent with the code. The naive A3 heuristic $n_{\text{eff}} = \lambda s$ would predict a physical concentration time $1/(\lambda g^*)$ (from $\lambda s = 1/g^*$) — but this is an *overestimate*: the optimal filter reaches the floor Σ_∞ faster, on the scale $\sim \tau_E$ (OU-mean-reversion, § S4.1), and the operational scale is precisely $\tau_{\text{conc}} = 1/g^*$; the conservativeness of A3 does not affect the σ -invariance of the window (§ 7 item iv). In adiabatic time $u = \lambda s$ the upper bound of the transient is $\lambda \tau_{\text{conc}} = \lambda/g^* = 2/\epsilon_{\text{track}}$. The structural unattainability of the slope (the key of S1.5) rests on the σ -invariance of the log-width of the fit window W (S1.2), not on the coincidence of the powers of σ . In the transient window the posterior variance about the “current level” of the latent decays as $\Sigma_{\text{tr}}(s) \asymp 1/(\mathcal{I} n_{\text{eff}}(s)) = 1/(\mathcal{I} \lambda s)$ — the standard Bayesian concentration $1/(\mathcal{I} n)$ with effective $n = \lambda s$ [5]; asymptotically $\Sigma_{\text{tr}} \rightarrow 1/(\mathcal{I} n_{\text{eff}}^*) = \Sigma_\infty$ (the floor merges with the stationary lag).

S1.3 Quadratic expansion of the instantaneous excess loss

The instantaneous increment of the excess loss at step s is the log-likelihood ratio of the oracle and the learner:

$$\Delta L_{\text{excess}}(s) = \ln \frac{p(X_s \mid \theta_v(s))}{\hat{p}(X_s \mid \hat{\theta}_v(s))}.$$

Averaging over $X_s \sim p(\cdot \mid \theta_v(s))$ gives the KL divergence $D_{KL}(p(\cdot \mid \theta_v(s)) \parallel \hat{p}(\cdot \mid \hat{\theta}_v(s)))$. For a regular parametric model, at small tracking error $\delta\theta_v := \hat{\theta}_v(s) - \theta_v(s)$, the Taylor expansion of the KL to second order (the first order vanishes at the point $\theta_v(s)$):

$$\mathbb{E}[\Delta L_{\text{excess}}(s)] = \frac{1}{2} \mathcal{I}(\theta_v(s)) \mathbb{E}[\delta\theta_v^2] + O(\mathbb{E}|\delta\theta_v|^3) \quad (\text{S1.5})$$

— the standard quadratic expansion of the regret in the Fisher metric [5, 6, 2] ch. 11. The tracking error $\mathbb{E}[\delta\theta_v^2]$ decomposes into stationary and transient parts (S1.2 and S1.4):

$$\mathbb{E}[\delta\theta_v^2(s)] = \underbrace{\Sigma_\infty}_{\text{stationary}} + \underbrace{\Sigma_{\text{tr}}(s)}_{\text{transient}} = \sigma_0^2(1 + O(\epsilon_{\text{track}})) + \frac{1}{\mathcal{I}\lambda s}(1 + o(1)). \quad (\text{S1.6})$$

S1.4 Integration and derivation of the coefficient $K/2$

Primary route: the MI telescope (the coefficient $K/2$ is manifest, without multiplication by λ). Before the step-by-step integration, let us fix the coefficient directly — by the redundancy–MI identity [5, 6] in the form derived in [1] § S8.1: the expected

cumulative excess loss of the ideal Bayesian learner (with the *mixture* predictive density $\hat{p}(X_s | X_{<s}) = \int p(X_s | \theta) \pi(\theta | X_{<s}) d\theta$, not a plug-in point estimate) telescopes into the accumulated mutual information of the memory about the latent. Within (A3), we decompose it into two terms corresponding to the two parts of the tracking error (S1.6) — the stationary lag Σ_∞ and the transient concentration $\Sigma_{\text{tr}}(s)$:

$$\mathbb{E}[L_{\text{excess}}(t)] = \underbrace{\sum_{s \leq t} \frac{1}{2} \text{tr}(\mathcal{I} \Sigma_\infty)}_{= A t \text{ (stat. lag, S1.9)}} + \underbrace{I(M_{\leq t}; \theta_v(t))}_{\text{concentration (log) term}}. \quad (\text{S1.6a})$$

The logarithm carries *only* the second, concentration term — the mutual information of the memory about the *current* latent $\theta_v(t)$. For a K -dimensional Gaussian latent with diagonal prior $\Sigma_{\text{prior}} = \sigma_0^2 \text{Id}_K$ and a Kalman posterior contracted to the effective number of observations $n_{\text{eff}}(t)$ ($\Sigma_{\text{post}} = (\mathcal{I} n_{\text{eff}})^{-1}$), this is a Gaussian MI — a prefactor $\frac{1}{2} \ln \det$:

$$I(M_{\leq t}; \theta_v(t)) = \frac{1}{2} \ln \frac{\det \Sigma_{\text{prior}}}{\det \Sigma_{\text{post}}(t)} = \frac{1}{2} \ln \det(\mathcal{I} \Sigma_{\text{prior}} n_{\text{eff}}(t)) = \frac{K}{2} \ln n_{\text{eff}}(t) + C_0, \quad (\text{S1.6b})$$

where $C_0 = \frac{1}{2} \ln \det(\mathcal{I} \Sigma_{\text{prior}})$ does not depend on t (all the dependence on $\mathcal{I}$ goes here: $\det(\mathcal{I} \Sigma_{\text{prior}} n_{\text{eff}}) = n_{\text{eff}}^K \det(\mathcal{I} \Sigma_{\text{prior}})$). The coefficient $K/2$ here is **manifest** — the prefactor $\frac{1}{2} \ln \det$ of the Gaussian MI, multiplied by the dimension $K = \dim \theta_v$; it does *not* arise from integrating ds/s and does *not* carry a factor λ . The drift parameter λ enters **only** the argument of the logarithm, through $n_{\text{eff}}(t) \asymp \lambda t$ (S1.4); yet the realization of $K/2$ precisely as the coefficient of the *observed* $\ln t$ still requires the linearity $n_{\text{eff}} \propto t^1$ (A3) — the prefactor $\frac{1}{2} \ln \det$ is manifest, while the *exponent* of the argument is set by A3 (consistent with § 3 main: A3 fixes the value of the transient logarithm, not the fact of transience):

$$I(M_{\leq t}; \theta_v(t)) = \frac{K}{2} \ln(\lambda t) + C'_0. \quad (\text{S1.6c})$$

Transience and the plateau arise here automatically, without an external truncation of the integral: $n_{\text{eff}}(t) = \min(\lambda t, n_{\text{eff}}^*)$ (S1.3b–c) grows as λt on the concentration burst ($t \lesssim \tau_{\text{conc}}$) and saturates at the floor $n_{\text{eff}}^* = 1/g^*$ for $t \gg \tau_{\text{conc}}$, so that $\frac{K}{2} \ln n_{\text{eff}}(t)$ passes into the plateau $\frac{K}{2} \ln(1/g^*)$ (S1.8a) exactly when the posterior about the current $\theta_v(t)$ ceases to concentrate. This is what fixes the logarithmic coefficient of Theorem 1.

Cross-check — the coordinate-wise per-step route (Riccati), where the same $K/2$ is reproduced under the correct measure of integration. Substituting (S1.6) into (S1.5) and integrating over s from some s_0 (after the initial transient) **on the transient concentration window** $s_0 \leq s \lesssim \tau_{\text{conc}}$, where the transient variance is still decaying according to (S1.4):

$$L_{\text{excess}}^{(ij)}(t) = \underbrace{\int_{s_0}^t \frac{1}{2} \mathcal{I} \Sigma_{\infty} ds}_{\text{measure } ds} + \underbrace{\int \frac{1}{2} \mathcal{I} \Sigma_{\text{tr}} dn_{\text{eff}}}_{\text{measure } dn_{\text{eff}} = \lambda ds} + \text{const} = \underbrace{\frac{1}{2} \mathcal{I} \Sigma_{\infty} (t - s_0)}_{\text{linear } a t} + \underbrace{\frac{1}{2} \int_{s_0}^{\min(t, \tau_{\text{conc}})} \frac{ds}{s}}_{\text{logarithmic term (S1.8)}}. \quad (\text{S1.7})$$

The upper limit of the logarithmic integral is $\min(t, \tau_{\text{conc}})$, **not** t : the integrand $\Sigma_{\text{tr}}(s) \asymp 1/(\mathcal{I}\lambda s)$ holds only while $n_{\text{eff}}(s) \asymp \lambda s$ is growing (S1.4), that is, on the transient window $s \lesssim \tau_{\text{conc}}$. At $s \gtrsim \tau_{\text{conc}}$ the steady-state g^* saturates n_{eff} at the floor $1/g^*$, the transient variance reaches Σ_{∞} , and the integrand Σ_{tr} ceases to give a $1/s$ -contribution — it is absorbed by the stationary term Σ_{∞} . This is precisely the structural cause of the transience of the logarithm, built into the mechanism rather than an external caveat.

More precisely: the transient contribution to (S1.5) is $\frac{1}{2} \mathcal{I} \cdot \Sigma_{\text{tr}}(s) = \frac{1}{2} \mathcal{I} \cdot \frac{1}{\mathcal{I}\lambda s} = \frac{1}{2n_{\text{eff}}(s)}$ (Fisher cancels, $n_{\text{eff}} = \lambda s$) — this is the excess loss *per single effective (decorrelated) observation*, i.e. the density $dL/dn_{\text{eff}} = 1/(2n_{\text{eff}})$, and *not* dL/ds : its natural argument is the number of effective observations n_{eff} , not the physical step s (the same $1/(2n_{\text{eff}})$ is the increment of $\frac{1}{2} \ln n_{\text{eff}}$ with respect to n_{eff} — BNT class II). The passage to physical time carries the *Jacobian of the effective-observation measure* $dn_{\text{eff}} = \lambda ds$ (one decorrelated observation per correlation interval $\tau_E = 1/\lambda$), so that the physical increment $\frac{dL}{ds} = \frac{dL}{dn_{\text{eff}}} \frac{dn_{\text{eff}}}{ds} = \frac{1}{2n_{\text{eff}}} \cdot \lambda = \frac{1}{2s}$ — exactly the conditional MI $I(X_s; \theta_v \mid X_{<s})$ of the telescope (S8.1) [1], yielding, summed over the K coordinates, $\dot{L}_{\text{excess}} \sim K/2t$ (S8.3) [1] (this is precisely what the direct step-by-step estimator of the `ou_adiaab` simulation § S4.1 measures: the static slope anchor 0.99). Naive integration of $\frac{1}{2} \mathcal{I} \Sigma_{\text{tr}}(s)$ *over physical* ds , by contrast, would give $\frac{1}{2\lambda} \ln$ — an overestimate by $1/\lambda$ from substituting the measure. The reduction to *adiabatic* time $u = \lambda s$ (the number of correlation intervals) motivates the *value* of the Jacobian — the passage to $n_{\text{eff}} \asymp \lambda s$ is *heuristically motivated* by the analogy with the Bartlett effective sample size $n_{\text{eff}} = n/(1 + 2 \sum_s \rho(s))$ [7] for observations with exponential autocorrelation $\rho(s) = e^{-\lambda|s|}$, relying on assumption (A3) about class-II concentration. *Carefully on the factor $\frac{1}{2}$* : for $\rho(s) = e^{-\lambda|s|}$ the sum of autocorrelations $1 + 2 \sum_{s \geq 1} \rho(s) \approx 2 \sum_{s \geq 0} e^{-\lambda s} \approx 2/\lambda$ for $\lambda \ll 1$, whence $n_{\text{eff}} = n/(1 + 2 \sum_s \rho) \approx n/(2/\lambda) = (\lambda/2)n$ — that is, with a factor $\frac{1}{2}$, $n_{\text{eff}} \approx (\lambda/2)n$, and *not* λn . This factor $\frac{1}{2}$ does not vanish without a trace: it enters only the *additive logarithmic constant*, not the slope, since $\ln((\lambda/2)s) = \ln(\lambda s) - \ln 2$ — the shift $-\ln 2$ per coordinate is absorbed by the constant c_{ij} (and further C) of formula (S1.9)–(S1.10), while the coefficient of $\ln(\lambda s)$ remains $\frac{1}{2}$. We therefore write $n_{\text{eff}} \asymp \lambda n$ *as an order-exact relation up to a multiplicative constant* $O(1)$ (here $\frac{1}{2}$), fixing only the *exponent* λs and hence the slope $K/2$; the exact value of the constant affects only C . This analogy was established by Bartlett for estimating the *mean* of a stationary correlated process; its formal transfer to *tracking* a drifting latent is not rigorously justified and is deferred to the open questions (§ 7 main). This is a *heuristic reduction to class II* [8], not an independent derivation; a rigorous control of the transient non-stationary

filter (the non-stationary Riccati equation) is deferred to the open questions § 7 main. The coefficient $1/2$ -per-parameter itself is the standard asymptotics of Bayesian / MDL universal coding [5, 6, 9], while its drift-scale-shifted predictive-information form is class II [8]. With this caveat, the loss per effective observation on a single coordinate is $\frac{1}{2n_{\text{eff}}}$, and its integral over dn_{eff} gives $\frac{1}{2} \ln n_{\text{eff}}$ — the same $\frac{1}{2}$ -per-coordinate as in (S1.6b). Rewriting the integral in physical time via the Jacobian $dn_{\text{eff}} = \lambda ds$, **on the transient window** $[s_0, \min(t, \tau_{\text{conc}})]$:

$$\int_{s_0}^{\min(t, \tau_{\text{conc}})} \underbrace{\frac{1}{2n_{\text{eff}}(s)}}_{dL/dn_{\text{eff}}} \underbrace{\lambda ds}_{dn_{\text{eff}}} = \int \frac{dn_{\text{eff}}}{2n_{\text{eff}}} = \frac{1}{2} \int_{s_0}^{\min(t, \tau_{\text{conc}})} \frac{ds}{s} = \frac{1}{2} \ln \frac{\min(t, \tau_{\text{conc}})}{s_0}. \quad (\text{S1.8})$$

Two regimes according to the relation between t and τ_{conc} . (i) *In the transient window* $s_0 \ll t \lesssim \tau_{\text{conc}}$ the integral gives $\frac{1}{2} \ln(t/s_0) = \frac{1}{2} \ln(\lambda t) - \frac{1}{2} \ln(\lambda s_0)$ — logarithmic growth with coefficient $\frac{1}{2}$ per coordinate, exactly class II [8]. *It is this regime* that fixes the analytic identity $B = K/2$ (the limit $\epsilon_{\text{track}} \rightarrow 0$). (ii) *Asymptotically*, for $t \gg \tau_{\text{conc}}$ (steady-state drift), the integral is truncated from above at τ_{conc} and the logarithmic term *passes into a constant plateau*

$$\frac{1}{2} \ln \frac{\tau_{\text{conc}}}{s_0} \asymp \frac{1}{2} \ln(1/g^*) \quad (t \gg \tau_{\text{conc}}), \quad (\text{S1.8a})$$

that is, per coordinate a constant $\frac{1}{2} \ln(1/g^*)$, and summed over the K coordinates a plateau of height $\sim (K/2) \ln(1/g^*)$. This removes the apparent contradiction between $n_{\text{eff}} \asymp \lambda t \rightarrow \infty$ (S1.4) and the saturation of n_{eff} at the floor: the former is the *transient* concentration burst ($s \lesssim \tau_{\text{conc}}$), the latter is the *asymptotics* of the steady-state drift ($s \gtrsim \tau_{\text{conc}}$); they pertain to different windows of the same integral, not to two incompatible premises.

The key point is the *cancellation of the Fisher information*: the transient variance $\Sigma_{\text{tr}}(s) = 1/(\mathcal{I}\lambda s)$ contains $1/\mathcal{I}$, while the quadratic factor (S1.5) contains $\mathcal{I}$; their product $\frac{1}{2}\mathcal{I} \cdot 1/(\mathcal{I}\lambda s) = 1/(2\lambda s)$ does not depend on $\mathcal{I}$. Therefore the logarithmic coefficient is determined *by the dimension alone*, and not by the curvature of the model — this is precisely the universality of class II [8]. Collecting (S1.7)–(S1.8) for a single coordinate *in the transient window* $s_0 \ll t \lesssim \tau_{\text{conc}}$:

$$L_{\text{excess}}^{(ij)}(t) = a_{ij} t + \frac{1}{2} \ln(\lambda t) + c_{ij} + o(1), \quad a_{ij} = \frac{1}{2} \mathcal{I}_{ij} \Sigma_{\infty}^{(ij)} = \frac{1}{2} \mathcal{I}_{ij} \sigma_0^2 (1 + O(\epsilon_{\text{track}})). \quad (\text{S1.9})$$

Summing over the K coordinates via (S1.1):

$$L_{\text{excess}}(t) = \underbrace{\left(\sum_{ij} a_{ij} \right)}_{=:A} t + \frac{K}{2} \ln(\lambda t) + C + o(1). \quad (\text{S1.10})$$

The logarithmic coefficient equals $\sum_{ij} \frac{1}{2} = K/2$ — identically, independently of σ and of the values $\mathcal{I}_{ij}$. The linear coefficient $A = \sum_{ij} \frac{1}{2} \mathcal{I}_{ij} \sigma_0^2 (1 + O(\epsilon_{\text{track}})) = \frac{1}{2} \sigma_0^2 \text{tr} \mathcal{I} \cdot (1 + O(\epsilon_{\text{track}})) \propto \sigma_0^2 = \sigma^2 / (2\lambda) \rightarrow 0$ as $\epsilon \rightarrow 0$. This proves the transient form (3.1) main (on the window $1 \ll \lambda t \lesssim \lambda \tau_{\text{conc}} = 2/\epsilon_{\text{track}}$); the asymptotic form (3.1') main — the line-plus-plateau $L_{\text{excess}}(t) = At + (K/2) \ln(1/g^*) + o(t)$ for $t \gg \tau_{\text{conc}}$ — is given by the truncation of the logarithmic integral at τ_{conc} according to (S1.8a). Both forms are equal parts of Theorem 1, not “an asymptotics and a corollary”. $\square$

Corollary (3.2) main. The meaningful scale of the undertow rate is not the referencing of L_{excess} to some energy budget, but the *excess loss per effective observation* — the growth of L_{excess} relative to the effective number of observations n_{eff} (§ 2.3, § 2.5b main). On the transient window $1 \ll \lambda t \lesssim \lambda \tau_{\text{conc}}$, where $n_{\text{eff}}(s) \asymp \lambda s$ is still growing (S1.4), and the logarithmic integral (S1.8) is truncated from above at $\min(t, \tau_{\text{conc}})$ according to (S1.8a), for $L_{\text{excess}}(t) \sim (K/2) \ln(\lambda t)$ (the leading term as $A \rightarrow 0$) this quantity behaves as

$$\frac{L_{\text{excess}}(t)}{n_{\text{eff}}(t)} \sim \frac{(K/2) \ln(\lambda t)}{\lambda t}, \quad \frac{d}{dt} \frac{(K/2) \ln(\lambda t)}{\lambda t} = \frac{(K/2)}{\lambda t^2} - \frac{(K/2) \ln(\lambda t)}{\lambda t^2} \sim -\frac{K \ln(\lambda t)}{2\lambda t^2},$$

where to leading order $\ln(\lambda t) \gg 1$ the second term dominates — the logarithmic contribution per observation decays to zero. For $t \gg \tau_{\text{conc}}$ the numerator reaches the plateau $(K/2) \ln(1/g^*)$ (S1.8a), the denominator reaches the floor $n_{\text{eff}}^* = (1 - g^*)/g^*$ (S1.3b), and the rate of logarithmic growth per observation vanishes. We emphasize: n_{eff}^* is an *informational* scale (the effective-observation floor under decorrelation, § 2.5b main), not an energetic denominator of efficiency. This is (3.2) main. $\square$

S1.5 Finite-amplitude correction and the convergence $\hat{B} \rightarrow K/2$

The true logarithmic coefficient equals $K/2$ (S1.10). The scan § S8.3 [1] observed $\hat{B}/(K/2) = 0.54, 0.78, 0.85$ at $\epsilon = 10^{-1}, 9 \cdot 10^{-3}, 10^{-3}$ — a shortfall decreasing monotonically with ϵ . We show that this is a bias of the *estimate* on a finite fit window, not a property of the true coefficient.

The joint fit $L_{\text{excess}}(t) \approx \hat{A}t + \hat{B} \ln(\lambda t) + \hat{C}$ on the window $[t_{\min}, t_{\max}]$ by least squares estimates $\hat{B}$ through the projection of the data onto the regressor $\ln(\lambda t)$, orthogonalized against $\{t, 1\}$. At finite ϵ the linear term At with $A \propto \epsilon$ is large (the drift fraction in the scan: 76% at $\epsilon = 10^{-1}$), and the regressors t and $\ln(\lambda t)$ on the finite window $[t_{\min}, t_{\max}]$ are statistically correlated (both grow monotonically). Incomplete orthogonalization transfers part of the variance of the logarithmic signal into the linear regressor, underestimating $\hat{B}$. The standard analysis of OLS bias with correlated regressors gives

$$\hat{B} = \frac{K}{2} (1 - g(\epsilon, W)), \quad g(\epsilon, W) = \frac{A \cdot \text{Cov}_W(t, \ell)}{(K/2) \cdot \text{Var}_W(\ell)} \Big|_{\ell=\ln(\lambda t)} = O(\epsilon \cdot \Theta(W)), \quad (\text{S1.11})$$

where $W = t_{\max}/t_{\min}$ is the ratio of the window bounds, $\Theta(W)$ is the regressor-overlap factor (increasing with the window width), $A \propto \epsilon$. Since $g \propto A \propto \epsilon$, we have $g(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$, and $\hat{B} \rightarrow K/2$. Qualitatively: $\hat{B}/(K/2)$ increases monotonically as ϵ decreases — 0.54 $\rightarrow$ 0.78 $\rightarrow$ 0.85 at $\epsilon : 10^{-1} \rightarrow 9 \cdot 10^{-3} \rightarrow 10^{-3}$ (that is, $g(\epsilon) = 0.46, 0.22, 0.15$ decreasing to zero) — which is exactly what $g(\epsilon) \rightarrow 0$ predicts. *The exact functional form of $g(\epsilon)$ is left open.* The theoretical dependence from (S1.11) is linear, $g \propto \epsilon^1$ (through $A \propto \epsilon$); the numerical fit $g \approx 1.5 \epsilon^{0.25}$ over three points, however, yields an exponent ≈ 0.25 , inconsistent with the theoretical 1. This exponent is *not derived* and is given only as a rough interpolation over three points of the *suboptimal* Robbins–Monro scan § S8.3 [1] (the very learner rejected below as a “negative control”); it is *not* a law and should not be read as one. The discrepancy of the exponents (theory ϵ^1 vs. fit $\epsilon^{0.25}$) we note as open — it probably reflects an additional $1/\sqrt{t}$ -bias of the suboptimal RM learner on top of the collinear bias (S1.11), rather than the true form of g ; the exact form requires an extended scan with the optimal filter. The load-bearing claim of S1.5 does not depend on the exponent: for it, a monotone $g(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ suffices, which is confirmed by all three points. As $\epsilon \rightarrow 0$ (the drift fraction $\rightarrow 0$) the linear term disappears, the regressors separate, and $\hat{B} \rightarrow K/2$ — which is precisely the analytic identity of Theorem 1.

Negative control § S8.3 [1]. Widening the window to $[10^3, T_{\text{total}}]$ gave $\hat{B}/(K/2) \rightarrow 2.0$ — an *overestimate*. By (S1.11) widening W increases $\Theta(W)$, but with the Robbins–Monro learner used, with step $1/\sqrt{t}$, on long horizons the learner’s *own* bias is added (suboptimality of the filter), introducing a spurious logarithmic signal. The EWMA filter with optimal forgetting scale $\tau_f^{\text{forget}} \asymp \tau_E$ (ou_adiab § S4.1) eliminates precisely this *overestimate*: for the optimal filter $\Sigma(t) \rightarrow \Sigma_\infty$ without a $1/\sqrt{t}$ -bias, and widening the window does not degenerate the fit upward. However, replacing the learner did *not* lead to attaining the limit $\hat{B} \rightarrow K/2$ as an asymptotic slope: the extended check § S4.1 with the *optimal* Kalman learner establishes that the logarithm is realized only transiently, while in a wide asymptotic window $\hat{B}/(K/2) \rightarrow 0$ (plateau), so that the elimination of the overestimate was confirmed, but convergence to the limit was not, and it is *structurally unattainable* (see below). $\square$

Reconciliation with the ou_adiab run § S4.1 (structural unattainability). The extended check § S4.1 with the *optimal* Kalman learner in the exact linear-Gaussian setting gives, in a wide asymptotic window ($W \approx 156\text{--}166$), $\hat{B}/(K/2)$ over 10 logarithmically uniform values of ϵ_{track} from $3 \cdot 10^{-1}$ to $3 \cdot 10^{-2}$ within $[-0.31; 0.16]$ around zero (9 of 10 bootstrap CIs cover 0, $R^2 = 1.000$): the series is flat around 0, not rising toward 1. This establishes that the working hypothesis “ $\epsilon_{\text{track}} = O(1)$, push $\epsilon_{\text{track}} \rightarrow 0$ — and $\hat{B} \rightarrow K/2$ ” is **false**. The cause is two-layered. The surface layer is two finite-window mechanisms, both underestimating $\hat{B}$: (a) the estimator bias (S1.11) $g(\epsilon) > 0$ under collinearity of t and $\ln(\lambda t)$; (b) the distribution of the loss — a well-tuned filter with fixed floor Σ_∞ assigns a larger fraction of the loss to the linear term At . The deep layer is *structural*: under OU drift the BNT class-II logarithm [8] accumulates only while the posterior concentrates ($n_{\text{eff}} \sim \lambda s$, transient burst $s \lesssim \tau_{\text{conc}}$,

S1.4); thereafter the steady-state g^* bounds n_{eff} at the floor $(1 - g^*)/g^* \approx 1/g^*$ (S1.3b), L_{excess} reaches a plateau of height $\sim (K/2) \ln(1/g^*)$ (S1.8a), and in any window $t \gg \tau_{\text{conc}}$ there is no logarithmic signal. The logarithmic width of the *asymptotic (plateau) fit window* $W = t_{\text{max}}/t_{\text{min}} = (c_0/\sigma^2)/(3/g^*) = c_0/(6\lambda r) \asymp \mathcal{I}_{\text{rate}}/\lambda = 1/(\lambda r)$ is **σ -invariant** (derivation — S1.2, “Why slope convergence is not achieved”): σ cancels between the lower bound $t_{\text{min}} = 3/g^* \propto 1/\sigma^2$ and the ceiling $c_0/\sigma^2 \propto 1/\sigma^2$, so that the limit $\epsilon_{\text{track}} \rightarrow 0$ stretches both bounds in the same proportion and the fit window stays in the plateau. Therefore the log-bearing transient ($t \lesssim \tau_{\text{conc}}$) and the asymptotic fit window (flat, $t \gg \tau_{\text{conc}}$) remain *separated* for all σ — and this is a consequence of the *adiabatic restriction* (fixed Fisher-rate per correlation interval, $\mathcal{I}_{\text{rate}}/\lambda = O(1)$), not of an arithmetic coincidence of the powers of σ : one can widen W only by increasing $\mathcal{I}_{\text{rate}}/\lambda \rightarrow \infty$, which takes one out of the slow-drift regime where (A1) and the quasi-stationarity of θ_v hold. Therefore the numerical convergence $\hat{B} \rightarrow K/2$ as an asymptotic slope is **structurally unattainable for a learner with saturating memory** (the optimal stationary filter). A learner with *growing* memory, for which n_{eff} relative to the current $\theta_v(t)$ does not saturate at the floor, *does not exist* in the linear-Gaussian window (A1): the floor $(1 - g^*)/g^*$ is a *provable lower bound* on n_{eff} over all memory architectures, not the limit of a particular one (see S1.6: sufficiency of the Kalman posterior + MMSE + DPI), so that storing an ever-larger number of raw old observations does not raise n_{eff} relative to $\theta_v(t)$. The decorrelation of old observations (the mechanism of Theorem 2) is an illustrative parallel, not the load-bearing argument. The loophole is closed by the optimality of the floor, not by the C3 expansion cost (§ S1.6; § 7 main, item v; § 8 main, “Mutual interlock”). The analytic identity $B = K/2$ is meanwhile confirmed by a separate static anchor (slope 0.990) and a transient burst (0.936). This is a single explanation, not competing ones.

S1.6 Optimality of the floor as a lower bound (sufficiency of the Kalman posterior and MMSE)

The thesis “the floor $n_{\text{eff}}^* = (1 - g^*)/g^*$ is not exceeded by any memory architecture”, on which the mutual interlock of the routes (§ 8 main) and the closing of the growing-memory loophole in S1.5 rest, is not postulated but proved — *in the linear-Gaussian window (A1)*. The proof consists of two standard facts of filtering theory, joined by the data-processing inequality.

Fact 1 (sufficient statistic). In the linear-Gaussian model (A1) — hidden OU state $\theta_v(t)$, Gaussian observations $Y_{1:t}$ — the posterior distribution $\theta_v(t) \mid Y_{1:t}$ is Gaussian and is fully specified by the pair $(\hat{\theta}_t, \Sigma_t)$, where $\hat{\theta}_t$ is the Kalman posterior mean and Σ_t the posterior covariance. This pair is an *exact finite-dimensional sufficient statistic* for $\theta_v(t)$ given the observations $Y_{1:t}$: the posterior $p(\theta_v(t) \mid Y_{1:t}) = p(\theta_v(t) \mid \hat{\theta}_t, \Sigma_t)$, and no other function of $Y_{1:t}$ carries information about $\theta_v(t)$ beyond $(\hat{\theta}_t, \Sigma_t)$ (the standard property of the Kalman filter for linear-Gaussian systems [4, 3]). The stationary $\Sigma_\infty = \lim_t \Sigma_t$ (S1.2) is the Bayes-optimal (MMSE) tracking error for $\theta_v(t)$: the estimate $\hat{\theta}_t$ minimizes

$\mathbb{E}[(\hat{\theta}_t - \theta_v(t))^2]$ over *all* measurable functions of $Y_{1:t}$, and the stationary Wiener–Kalman–Bucy filter attains this minimum [4] ch. 4–5. In other words, Σ_∞ is not a property of a *particular* memory architecture, but a lower bound on the tracking error over *all* architectures.

Fact 2 (DPI for arbitrary memory). Let $S = f(Y_{1:t})$ be *any* memory (any measurable function of the observations: a recursive $O(1)$ -statistic, a FIFO buffer of growing size, the full log of raw observations). By the data-processing inequality

$$I(S; \theta_v(t)) \leq I(Y_{1:t}; \theta_v(t)),$$

and the right-hand side is attained by the sufficient statistic $(\hat{\theta}_t, \Sigma_t)$ of the optimal filter (Fact 1): $I((\hat{\theta}_t, \Sigma_t); \theta_v(t)) = I(Y_{1:t}; \theta_v(t))$. Consequently, no memory S carries more information about $\theta_v(t)$ than the Kalman posterior, and the MMSE error of any architecture is no smaller than the optimal one:

$$\mathbb{E}[(\mathbb{E}[\theta_v(t) | S] - \theta_v(t))^2] \geq \Sigma_\infty,$$

moreover, for finite t the lower bound is $\Sigma_t \geq \Sigma_\infty$ (monotone decrease of the posterior covariance toward the stationary limit), so $\geq \Sigma_\infty$ holds a fortiori; the floor n_{eff}^* corresponds to the stationary regime $t \gg \tau_{\text{conc}}$. For arbitrary memory $S = f(Y_{1:t})$ we set $n_{\text{eff}}(S) := 1/(\mathcal{I} \mathbb{E}[(\mathbb{E}[\theta_v(t) | S] - \theta_v(t))^2])$ — the effective number of observations, defined through the *achievable* tracking error; then $n_{\text{eff}}(S) \leq n_{\text{eff}}^*$ is a direct rewriting of $\mathbb{E}[\cdot] \geq \Sigma_\infty$, and the comparison of architectures by n_{eff} is legitimate without a separate assumption about geometric weighting (the identity $\Sigma_\infty = 1/(\mathcal{I} n_{\text{eff}}^*)$ fixes the correspondence). The effective number of informative observations n_{eff} of the present work is the number of past observations that the stationary filter weights with a geometric window (factor $(1 - g^*)$ per step), $n_{\text{eff}}^* = (1 - g^*)/g^* \approx 1/g^*$ (S1.3b): it is set by the steady-state g^* , which in turn is the *optimal* (MMSE-minimizing) balance of bias and variance for the drifting latent. Since no memory reduces the tracking error below Σ_∞ , no architecture realizes a *longer* effective averaging window relative to the *current* $\theta_v(t)$ than the optimal filter: lengthening the window (larger n_{eff} , smaller g^*) for a drifting target *increases* the lag error and violates the MMSE optimum. Therefore the floor n_{eff}^* is the *maximum* of the effective window over all memory architectures, not the limit of a particular (saturating) memory; raising n_{eff} relative to $\theta_v(t)$ above it, without violating the MMSE bound Σ_∞ , is impossible by any means. $\square$

Domain of applicability and connection with C2. The proof of finite-dimensionality of the sufficient statistic relies *essentially* on linear-Gaussianity (A1): only for a linear-Gaussian observation model does the posterior close on the finite-dimensional pair $(\hat{\theta}_t, \Sigma_t)$. For a nonlinear or non-Gaussian observation model the posterior $p(\theta_v(t) | Y_{1:t})$ is in general *not* finite-dimensional (exact filtering requires storing the entire density), and the MMSE optimality of the finite-dimensional floor is not guaranteed — this regime is open (§ 7 main, item iv). Within (A1) the argument is rigorous and does not rely on the decorrelation of old observations: the decorrelation $\beta_v(\Delta t) \rightarrow 0$ (the mechanism of

Theorem 2, C2) provides an *illustrative parallel* — “in old observations there is simply no needed information about $\theta_v(t)$ ” — but the *load-bearing* argument for the optimality of the floor is sufficiency + MMSE (Facts 1–2), not decorrelation. These two are one and the same consequence of the mixing OU structure, read from two sides, but for the lower bound Facts 1–2 suffice. The load-bearing object here is the pointwise $\theta_v(t)$, since the floor n_{eff} is defined through tracking the *current* latent; the trajectory version (horizon τ , condition (4.1)) is needed only by the C2 mechanism and is connected to the pointwise one for $\tau = O(1)$ (§ S2.0, Remark “trajectory versus pair”; § 8 main), but is not required for the lower bound on n_{eff} .

What the qualitative no-go carries and what the A3 heuristic carries (separating the load-bearing premises). Let us emphasize what exactly the no-go (Proposition 1 main) rests on, so that it does not appear to hinge on the heuristic reduction (A3, $n_{\text{eff}} \asymp \lambda s$). The *qualitative* no-go — that the logarithm is transient, that L_{excess} reaches a line-plus-plateau, and that the fitted slope $B \rightarrow 0$ in a wide asymptotic window — already follows from the **finiteness of the stationary tracking error** $\Sigma_\infty < \infty$, and this is a *rigorous* result of the algebraic Riccati equation (S1.2), not a heuristic. The chain is rigorous and does not use A3: $\Sigma_\infty < \infty$ (the positive Riccati root (S1.2)) $\Rightarrow$ the posterior variance about the *current* $\theta_v(t)$ is bounded below by the quantity $\Sigma_\infty > 0$ at all t (Facts 1–2: this is the MMSE optimum, not surpassed by any memory) $\Rightarrow$ the effective number of informative observations saturates at the floor $n_{\text{eff}}^* = (1 - g^*)/g^* < \infty$ (the identity $\Sigma_\infty = 1/(\mathcal{I} n_{\text{eff}}^*)$) $\Rightarrow$ the integral (S1.7)–(S1.8) of the logarithmic term is truncated from above at τ_{conc} , and the logarithm is transient (the plateau (S1.8a)). Each link rests on the finiteness of Σ_∞ and the optimality of the filter, *not* on $n_{\text{eff}} \asymp \lambda s$. The A3 heuristic is needed **only** for the *value* of the coefficient $K/2$ within the logarithmic burst (through $n_{\text{eff}} \asymp \lambda s$ on the transient window, S1.4). The *scale* $\tau_{\text{conc}} = n_{\text{eff}}^* = (1 - g^*)/g^* \approx 1/g^*$ itself, by contrast, is a strictly-Riccati quantity — it is set by the steady-state g^* (the Riccati root (S1.2)), *not* by A3. Neither transience as such, nor the existence of the plateau, nor the vanishing of the asymptotic slope depends on A3 — they follow from the rigorous Riccati ($\Sigma_\infty < \infty$) and S1.6 (the floor as the MMSE optimum). Therefore the load-bearing weight of the no-go lies on the *rigorous* Riccati plus the floor-optimum S1.6, while the A3 heuristic determines only the *value* of $K/2$ of the transient logarithm — even if A3 is quantitatively wrong, the no-go (transience) is preserved, and only the numerical coefficient within the burst changes.

S2. Proof of Theorem 2 (universality of the floor)

Theorem 2 main: for an admissible drift process (mixing condition (4.1) main) under FIFO, $c < 1$ and per-bit uniformity, $\liminf_{t \rightarrow \infty} \nu^{\text{theor}}(t) \geq 1 - c$ holds. Proof: (S2.0) general scheme via the mixing condition; (S2.1) PSP; (S2.2) fOU with dependence of the exponent on H ; (S2.3) multi-scale; concludes by verifying that OU is a special case.

S2.0 General scheme via the mixing condition

We reproduce the structure of [1] § 4.4, replacing the step “spectral gap of OU” with the general condition (4.1) main. The chain rule for mutual information:

$$I(M_{\leq t}; X_E^{[t, t+\tau]}) = I(M_{\leq t}^{\text{refr}}; X_E^{[t, t+\tau]}) + I(M_{\leq t}^{\text{stale}}; X_E^{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}}). \quad (\text{S2.1})$$

The stale fraction $M_{\leq t}^{\text{stale}}$ is fixed at the moment $s \leq t - \Delta t$ as a *deterministic function* of $\theta_v(s)$ (the writing of the bit at moment s). Crucially: the Markovianity of the drift is **not** used here — the argument relies only on the data-processing inequality for deterministic encoding and on the stationarity of increments. Since $M^{\text{stale}} = f(\theta_v(s))$, by DPI for the first link

$$I(M_{\leq t}^{\text{stale}}; X_E^{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}}) \leq I(\theta_v(s); X_E^{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}}).$$

Further, $X_E^{[t, t+\tau]}$ is a function of the trajectory $\theta_{[t, t+\tau]}$ (observations are generated by the current latent), so again by DPI

$$I(\theta_v(s); X_E^{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}}) \leq I(\theta_v(s); \theta_{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}}) \leq C \cdot \beta_v(\Delta t), \quad (\text{S2.2})$$

where the last inequality is exactly the mixing condition (4.1) main. Neither of the two DPI links requires Markovianity: the first — the determinism of the encoding M^{stale} , the second — that observations are a function of the latent trajectory. This is precisely why the argument goes through for *non-Markovian* fOU with $H > 1/2$ (the increments are correlated at all scales, there is no spectral gap) — this is exactly the substantive part of § 4: mixing suffices (a majorant $\beta_v(\Delta t)$ for $I(\theta_v(s); \theta_{[t, t+\tau]})$), not a Markov decomposition. This is a refinement, not a refutation of [1]: Markovianity in Lemma 2 is a *sufficient*, not a necessary condition, and Lemma 2 is correct within its OU class; Theorem 2 merely shows that the DPI chain goes through even without Markovianity.

Remark (trajectory versus pair). Condition (4.1) main majorizes the mutual information of a stale bit with the *entire future trajectory* of the latent $\theta_{[t, t+\tau]}$ over the prediction horizon, whereas the explicit $\beta_v(\Delta t)$ computed below for each class (S2.3–S2.6) are *pairwise* quantities $I(\theta_v(s); \theta_v(t))$. These two objects coincide in decay exponent in the classes considered, for two reasons. (i) For *Markovian* OU and PSP the supremum over the horizon τ does not change the exponent: by Markovianity $I(\theta_v(s); \theta_{[t, t+\tau]}) = I(\theta_v(s); \theta_v(t))$ — all dependence on the past passes through the current state $\theta_v(t)$, so the trajectory and pairwise majorants are identical. (ii) For *non-Markovian* fOU with $H \in (1/2, 3/4)$ the trajectory information is majorized by the *multiple* (block) correlation of $\theta_v(s)$ with the future block — **without** appeal to the monotonicity of conditioning. Rigorously (for $\tau = O(1)$, on which the proof is built): the pair “past sample $\theta_v(s)$ — future block $\Theta_v := (\theta_v(t), \dots, \theta_v(t + \tau))$ ” is jointly Gaussian (fOU is Gaussian), so the mutual information is expressed through the *square of the multiple correlation* $R^2 := R^2(\theta_v(s); \Theta_v)$ (the coefficient of determination of the linear

regression of $\theta_v(s)$ on the block Θ_v) in closed form

$$I(\theta_v(s); \theta_{[t, t+\tau]}) = -\frac{1}{2} \ln(1 - R^2) = \frac{1}{2} R^2 (1 + o(1)) \quad (R^2 \rightarrow 0).$$

Here $R^2 = \mathbf{r}^\top \Gamma^{-1} \mathbf{r} / \sigma_0^2$, where $\mathbf{r} = (\text{Cov}(\theta_v(s), \theta_v(u)))_{u \in [t, t+\tau]}$ is the vector of lag covariances, and Γ is the covariance matrix of the block Θ_v . This is a *nonnegative quadratic form in $\mathbf{r}$* (since $\Gamma^{-1} \succ 0$), and its decay exponent in $\Delta t = t - s$ is set by the leading component of $\mathbf{r}$: for $\Delta t \gg \tau$ all $\tau + 1$ components of $\mathbf{r}$ are of one order $\sigma_0^2 \rho_H(\Delta t)$ (by (S2.4), for $H \in (1/2, 3/4)$ the tail $\rho_H \sim s^{2H-2}$ monotonically *decreases* on the scale $\Delta t \gg \tau$, so $|r_u| = \sigma_0^2 \rho_H(\Delta t + u) \leq \sigma_0^2 \rho_H(\Delta t)$), whence $\|\mathbf{r}\|^2 \leq (\tau + 1) \sigma_0^4 \rho_H^2(\Delta t)$ and

$$R^2 \leq \frac{\|\mathbf{r}\|^2 \|\Gamma^{-1}\|}{\sigma_0^2} \leq C(\tau) \rho_H^2(\Delta t), \quad C(\tau) := \frac{(\tau + 1) \sigma_0^2}{\lambda_{\min}(\Gamma)} = O(1) \text{ for fixed } \tau = O(1),$$

where $\lambda_{\min}(\Gamma) > 0$ (positive definiteness of the block of the stationary fOU). Consequently $I(\theta_v(s); \theta_{[t, t+\tau]}) \leq \frac{1}{2} C(\tau) \rho_H^2(\Delta t) = C'(\tau) (\Delta t)^{-4(1-H)}$ — **the same leading exponent $4(1-H)$ as for the pairwise $\beta_v(\Delta t) = \frac{1}{2} \rho_H^2(\Delta t)$** (S2.5), where the factor $C(\tau)$ at fixed $\tau = O(1)$ is bounded and enters only the constant, not the exponent. (The spectral-norm majorant $\|\mathbf{r}\|^2 \|\Gamma^{-1}\|$ is taken only for transparency of the constant $C(\tau)$; the *exact* block $I(\theta_v(s); \Theta_v)$ decays with the same exponent $4(1-H)$, not faster — the ratio $R^2 / \rho_H^2(\Delta t)$ tends to a finite positive constant, since the leading lag component of $\mathbf{r}$ dominates.) No sign of partial correlations and no inequality “conditioning does not increase MI” is used here: the estimate proceeds directly through the multiple correlation of the jointly Gaussian block, which is valid also for non-Markovian fOU (where there is no monotonicity of MI under conditioning). For general *growing* τ the estimate is *not uniform* in τ : $C(\tau)$ *grows* with τ — both through the explicit factor $(\tau + 1)$ and through the decrease of $\lambda_{\min}(\Gamma)$ (which may tend to zero as the intra-block correlation of long-memory fOU strengthens with block growth), so the block R^2 is no longer majorized by a single lag component, and no rigorous statement is claimed — this case is honestly relegated to the open ones (§ 7 main, item iv). Thus the proof is built at horizon $\tau = O(1)$ (one-step predictive information, § 2.2 main), at which the trajectory majorant coincides with the pairwise β_v in exponent, and the computed pairwise β_v give the correct rate exponent.

From the fraction of fresh bits to the bound on I_{pred} (expanded step). The passage from the arithmetic fraction of non-refreshed bits to a majorant on the ratio of predictive informations proceeds in four explicit steps; we unfold it, since it is precisely here that per-bit uniformity is folded in and the correct renewal result is used.

- (a) *Decomposition of the contribution by bits.* By per-bit uniformity [1] Remark 5 the learner’s posterior factorizes over the independently refreshed coordinates (an explicit **ASSUMPTION of a factorized posterior**; for fOU/PSP/multi-scale it holds, since the coordinates are independent by construction, see S2.5 [and § 5 main]). *Caveat on the softmax observational coupling (parallel to A1 from C1).* Per-bit-uniformity is exact for the factorization of the *latent* coordinates θ_{ij} (the

increments dW_{ij} are independent by construction, S1.1). But the observable $P(t)$ is built by a *row-wise softmax*, coupling the coordinates of a row via a common normalization, so the learner's posterior over the observations does not factorize *exactly* — exactly the same situation as in C1, where the softmax linearization is relegated to assumption A1. This coupling introduces a correction of order $O(\sigma^2/\lambda) = O(\epsilon)$ to the uniformity of the per-bit contribution (the same order of accuracy as the remainder of the EKF/Laplace approximation A1 in § S1, controlled in the adiabatic limit $\epsilon \rightarrow 0$). Crucially for the lower bound: the correction is $O(\epsilon)$ *in either direction* — its sign (synergy versus redundancy) is in general *not* fixed, since $I(X_1, \dots, X_n; Y)$ versus $\sum_i I(X_i; Y)$ is sign-indefinite (the coupling structure may yield an excess as well). Heuristically, in the adiabatic near-Gaussian limit the positive correlation of the fresh bits of a row rather *decreases* their aggregate independent contribution relative to the factorized estimate $N_{\text{fresh}} \cdot I_{\text{bit}}^{\text{opt}}$ (the upper majorant in (S2.2a)–(b)), raising the floor — but this is a heuristic, not a proven sign. Rigorous control of this softmax correction (the exact non-uniformity of the per-bit contribution under coupled normalization) is open (§ 7 main, item iv); within the adiabatic approximation, as with A1 in C1, it is absorbed into $(1 + o(1))$. Therefore the total predictive information of the memory is additive over bits, and each retained bit contributes *uniformly* — of order $I_{\text{bit}}^{\text{opt}} := I_{\text{pred}}^{\text{opt}}/|M_{\leq t}|$ per bit if it is fresh, and $\leq C \cdot \beta_v(\text{age})$ if it is stale (by the DPI bound (S2.2)). Splitting the memory into fresh (age $\leq \tau_E$) and stale (age $> \tau_E$) fractions:

$$I_{\text{pred}}(t, \tau) \leq \underbrace{\sum_{\text{fresh}} I_{\text{bit}}^{\text{opt}}}_{\leq N_{\text{fresh}} \cdot I_{\text{bit}}^{\text{opt}}} + \underbrace{\sum_{\text{stale}} C \cdot \beta_v(\text{age})}_{\rightarrow 0}, \quad (\text{S2.2a})$$

where N_{fresh} is the number of fresh bits.

Caveat on the joint conditional MI of the stale fraction. The second sum in (S2.2a) is written as a *sum* of per-bit $\beta_v(\text{age})$, but the quantity it majorizes is the *joint conditional* mutual information of the set of stale bits $I(M_{\leq t}^{\text{stale}}, X_E^{[t, t+\tau]} \mid M_{\leq t}^{\text{refr}})$; the sum of per-bit β_v does not majorize the joint one without conditional independence of the stale bits, and for long-memory fOU with $H > 1/2$ there is none (the increments are correlated at all scales). The correct majorant of the joint conditional MI of the stale fraction is via the same block *multiple* correlation R^2 introduced for the trajectory majorant in the Remark above (S2.2): the joint information of the stale block with the future trajectory is estimated by the jointly Gaussian R^2 form $-\frac{1}{2} \ln(1 - R^2) = \frac{1}{2} R^2(1 + o(1))$, which gives the same leading exponent $4(1 - H)$, and *not* by the arithmetic sum of per-bit β_v . The correction from conditioning on $M_{\leq t}^{\text{refr}}$ (explaining-away — conditioning may *increase* MI) is of higher order in ρ_H : for Gaussian fOU, via the Schur complement, the correlation of $M_{\leq t}^{\text{refr}}$ with the stale $\theta_v(s)$ is itself $O(\rho_H(\Delta t))$, so its contribution to the conditional MI of the stale fraction is multiplicative in ρ_H and does not change the leading exponent $4(1 - H)$. The general non-Gaussian case (where the Schur complement

is inapplicable and the sign of explaining-away is not controlled) is honestly relegated to the open ones (§ 7 main, item iv).

- (b) *The first sum $\leq c \cdot I_{\text{pred}}^{\text{opt}}$ via per-bit uniformity.* Here what is needed is precisely the *uniformity of the per-bit contribution* (per-bit uniformity), not merely additivity of MI, and this distinction is essential. Additivity of mutual information over independent coordinates (S1.1, [2] ch. 2.5) gives only $I_{\text{pred}} = \sum_{\text{bits}} I_{\text{bit}}$ — it does *not* forbid fresh bits from carrying a disproportionately large share of $I_{\text{pred}}^{\text{opt}}$. We need something *stronger*: that the limiting contribution of each fresh bit does not exceed the *mean* $I_{\text{bit}}^{\text{opt}} = I_{\text{pred}}^{\text{opt}}/|M_{\leq t}|$. This uniformity is provided by the *symmetry* of the independent coordinates (under a factorized posterior and symmetric FIFO all coordinates are statistically interchangeable, so the limiting per-bit contribution is identical and equal to the mean), which, in turn, is guaranteed by additivity of MI *precisely for symmetric independent coordinates* (S1.1). That is, additivity is a *necessary ingredient* of uniformity, but is not identical to it: per-bit uniformity = additivity + symmetry of coordinates. Since then each fresh bit contributes no more than $I_{\text{bit}}^{\text{opt}} = I_{\text{pred}}^{\text{opt}}/|M_{\leq t}|$, the first sum is majorized by $N_{\text{fresh}} \cdot I_{\text{pred}}^{\text{opt}}/|M_{\leq t}| = (N_{\text{fresh}}/|M_{\leq t}|) \cdot I_{\text{pred}}^{\text{opt}}$. The fraction of fresh bits $N_{\text{fresh}}/|M_{\leq t}|$ is exactly $c(1+o(1))$ — this is established in step (c). Thereby $\sum_{\text{fresh}} \leq c(1+o(1)) \cdot I_{\text{pred}}^{\text{opt}}$. *Domain of validity of this step (essential for C3, § S3.2).* The uniformity of the per-bit contribution carries *inter-coordinate* symmetry (each of the K coordinates is estimated once), but **not** *inter-age* uniformity: the majorant $N_{\text{fresh}} \cdot I_{\text{bit}}^{\text{opt}}$ silently assumes that several fresh snapshots of *one* coordinate of different ages do not add up beyond the mean. This is true in the natural regime of C2 — $N_{\text{fresh}}/K < n_{\text{eff}}^*$ (no more than one effective fresh observation per coordinate; with fixed memory $|M_{\leq t}| = O(K n_{\text{eff}}^*)$ this is indeed so). If, however, the memory is so large that $N_{\text{fresh}}/K \geq n_{\text{eff}}^*$, the fresh snapshots *jointly* estimate $\theta_v(t)$ down to the tracking floor Σ_{∞} (§ S1.6), the first sum ceases to be majorized by $c \cdot I_{\text{pred}}^{\text{opt}}$, and the floor drops from $1-c$ to the C1 floor ν_{C1} — it is precisely this mechanism that carries Theorem 3 under growing capacity (§ S3.2).
- (c) *The fraction of fresh bits = c: the stationary age distribution of the FIFO buffer.* Here what is required is *not* the elementary renewal theorem (it is about $\mathbb{E}[N(t)]/t \rightarrow 1/\mathbb{E}[\text{interval}]$ — growth of the *number* of events), but the *stationary age distribution* in a buffer of fixed size under growing influx (backward recurrence time, [10] ch. 5). *Terminological refinement:* for *deterministic* FIFO the survival time of a bit is constant (exactly $|M_{\leq t}|/\dot{M}_{\text{refresh}}$ for each written bit), so the age distribution is *uniform without size-biasing* — the classical inspection-paradox skew (weighting of long intervals) arises only for *random* interval lengths, whereas here the survival interval is deterministic and identical, so the residual time is uniformly distributed trivially. That is, we use Cox’s backward-recurrence picture, but *without* its size-biased component — the determinism of FIFO removes it. For a FIFO buffer of capacity $|M_{\leq t}|$, into which bits are written at rate $\dot{M}_{\text{refresh}}$, a bit-snapshot of age

a is present if and only if it was written in the window $[t - a, t]$ and not yet evicted; in the steady-state regime the age density is uniform on $[0, |M_{\leq t}|/\dot{M}_{\text{refresh}}]$ (each written bit “survives” until eviction over time $|M_{\leq t}|/\dot{M}_{\text{refresh}}$). The fraction of bits of age $\leq \tau_E$ is then the ratio of the freshness window to the total survival time:

$$\frac{N_{\text{fresh}}}{|M_{\leq t}|} = \frac{\tau_E}{|M_{\leq t}|/\dot{M}_{\text{refresh}}} = \frac{\dot{M}_{\text{refresh}} \tau_E}{|M_{\leq t}|} = c, \quad (\text{S2.2b})$$

exactly the definition of c . The correctness of the fraction (≤ 1) and the uniformity of the age distribution over the *full* survival time $|M_{\leq t}|/\dot{M}_{\text{refresh}}$ are guaranteed by the hypothesis $c < 1$ of Theorem 2: for $c \geq 1$ we would have $\dot{M}_{\text{refresh}} \tau_E \geq |M_{\leq t}|$ (the influx of fresh bits over the window τ_E fills the buffer entirely), the fresh fraction saturates at 1, the survival window collapses to τ_E , and the uniform distribution on $[0, |M_{\leq t}|/\dot{M}_{\text{refresh}}]$ ceases to be correct; $c < 1$ is precisely what excludes this degeneracy. (Under growing influx this “locally stationary” distribution — uniformity holds on the scale $\ll |M_{\leq t}|/\dot{M}_{\text{refresh}}$ on which the influx is constant; the adiabatic refinement for growing $|M_{\leq t}|$ is § S3.2.) Accordingly, the fraction of stale bits is $1 - c$.

(d) *The second sum* $\rightarrow 0$. The aggregate contribution of the stale fraction, averaged over the uniform age distribution, is majorized by $\int_{\tau_E}^{\infty} C \beta_v(a) \frac{da}{|M_{\leq t}|/\dot{M}_{\text{refresh}}}$; for $\int_0^{\infty} \beta_v ds < \infty$ (i.e. $H < 3/4$ for fOU, S2.2; exponential β_v for OU/PSP/multi-scale) the integral is finite, and normalization by the growing survival time $|M_{\leq t}|/\dot{M}_{\text{refresh}}$ nullifies the contribution up to $O(\beta_v(\tau_E)) = o(1)$ under slow drift $\beta_v(\tau_E) \ll 1$. This is precisely the condition $\int_0^{\infty} \beta_v ds < \infty$ that makes the DPI majorant of the stale fraction summable, not merely pointwise-decaying.

Collecting steps (a)–(d) — (S2.2a), (S2.2b) and the estimates of the sums: $I_{\text{pred}}(t, \tau) \leq c(1 + o(1)) \cdot I_{\text{pred}}^{\text{opt}}(t, \tau) + o(I_{\text{pred}}^{\text{opt}})$, whence

$$\frac{I_{\text{pred}}(t, \tau)}{I_{\text{pred}}^{\text{opt}}(t, \tau)} \leq c \cdot (1 + o(1)) \implies \nu^{\text{theor}}(t) = 1 - \frac{I_{\text{pred}}}{I_{\text{pred}}^{\text{opt}}} \geq 1 - c \cdot (1 + o(1)),$$

whence $\liminf \nu^{\text{theor}}(t) \geq 1 - c$. This is (4.2) main. It remains to verify that each of the three non-OU classes satisfies (4.1) main, and to establish the *rate* via the explicit form of $\beta_v(\Delta t)$. $\square$ (general scheme)

S2.1 PSP (Poisson reset)

The PSP logit $\theta_v(t)$ is piecewise constant; the jump points form a Poisson stream of intensity $\mu = 1/\tau_E$; on each interval θ_v is independently redrawn. The probability that there was no jump between s and t is $e^{-\mu(t-s)} = e^{-(t-s)/\tau_E}$; upon a jump $\theta_v(t)$ is independent of $\theta_v(s)$. The mixing condition (4.1) and the DPI chain S2.0 need a majorant on the *observational* information $I(\theta_v(s); X_E^{[t, t+\tau]})$, and **not** on the latent-latent one $I(\theta_v(s); \theta_v(t))$: for piecewise-constant PSP, in the absence of a jump $\theta_v(t) = \theta_v(s)$

exactly, so the joint law of the pair is singular (an atom on the diagonal relative to the product of the marginals) and $I(\theta_v(s); \theta_v(t)) = +\infty$ — the latent quantity is unusable as a finite majorant. Conditioning on the jump indicator J on $(s, t]$ (the Poisson stream is independent of the redrawn values, $I(\theta_v(s); J) = 0$): upon a jump (probability $1 - e^{-\mu\Delta}$) the future $X_E^{[t, t+\tau]}$ is independent of $\theta_v(s)$ — the contribution is zero; in the absence of a jump (probability $e^{-\mu\Delta}$) the contribution is bounded by the *finite* predictable capacity of the horizon $H_\tau := H(X_E^{[t, t+\tau]}) \leq (\tau + 1) \ln k < \infty$. Hence

$$I(\theta_v(s); X_E^{[t, t+\tau]}) \leq e^{-\mu\Delta} H_\tau =: \beta_{\text{PSP}}(\Delta t) = H_\tau e^{-\Delta t/\tau_E}. \quad (\text{S2.3})$$

Key point: the majorant is built on the *finite observational* H_τ (not on the differential entropy of the latent H_{θ_v} , which for a continuous latent with an exact copy yields $I = +\infty$), and the DPI chain S2.0 for PSP closes on the finite $I(\theta_v(s); X_E^{[t, t+\tau]})$, bypassing the divergent latent-latent quantity.

Mixing is *exponential* with scale $\tau_E = 1/\mu$. Condition (4.1) is satisfied; $\int_0^\infty \beta_{\text{PSP}} ds = H_\tau \tau_E < \infty$. The floor constant $1 - c$ depends on μ only through τ_E in c ; the rate of approach is exponential with the same scale. The more frequent the switches (larger μ , smaller τ_E), the faster the decorrelation, but also the smaller c at fixed $\dot{M}_{\text{refresh}}$ and $|M_{\leq t}|$, that is, the *higher* the floor $1 - c$. This agrees with intuition: a rapidly resetting environment devalues memory faster. $\square$

S2.2 fOU (fractional OU; dependence of the exponent on H)

fOU is the stationary Gaussian solution of the Langevin equation with fractional Brownian forcing B^H , $H \in (0, 1)$ [11, 12]. The autocorrelation of the stationary fOU at large lags:

$$\rho_H(s) = \mathbb{E}[\theta_v(t)\theta_v(t+s)]/\sigma_0^2 \sim \begin{cases} e^{-s/\tau_E}, & H = 1/2 \text{ (degeneration into OU)}, \\ H(2H - 1)(s/\tau_E)^{2H-2}, & H > 1/2 \text{ (long memory)}, \\ H(2H - 1)(s/\tau_E)^{2H-2}, & H < 1/2 \text{ (antipersistence)}. \end{cases} \quad (\text{S2.4})$$

For the Gaussian pair $(\theta_v(s), \theta_v(t))$ the mutual information $I = -\frac{1}{2} \ln(1 - \rho_H^2) = \frac{1}{2} \rho_H^2(1 + o(1))$ as $\rho_H \rightarrow 0$ ([1] § S1.5, formula (S1.5a)). Substituting (S2.4) for $H > 1/2$:

$$\beta_{\text{fOU}}(\Delta t) = \frac{1}{2} \rho_H^2(\Delta t) \sim \frac{1}{2} [H(2H - 1)]^2 (\Delta t/\tau_E)^{2(2H-2)} = C_H \cdot (\Delta t)^{-(4-4H)}, \quad (\text{S2.5})$$

— *power-law* mixing with exponent $4 - 4H = 4(1 - H)$. *Refinement on the regime change at the point $H = 1/2$.* The exponent $4(1 - H)$ characterizes the *power-law* regime $H > 1/2$; the formal substitution $H \rightarrow 1/2$ gives $4(1 - H) \rightarrow 2$, but this is *not* the transition limit: at $H = 1/2$ the autocorrelation (S2.4, first row) changes class to *exponential* ($\rho_H \sim e^{-s/\tau_E}$, not the power-law Δt^{-2}), and $\beta_v \sim e^{-2\Delta t/\tau_E}$ decays faster than

any power. Therefore $\Delta t^{-4(1-H)}$ as $H \rightarrow 1/2^+$ is not the transition limit to OU, but the *boundary of the domain of applicability* of the power-law form (S2.5): the power-law and exponential regimes are different decay classes, separated by the point $H = 1/2$, and not connected by a continuous limiting transition in the exponent. The sufficient condition (4.1) main in its rigorous form requires convergence of the integral $\int_0^\infty \beta_{\text{fOU}} ds < \infty$, that is $4(1-H) > 1$, or $H < 3/4$ — precisely in this range the summation of contributions of stale bits over the uniform age distribution is majorized by a convergent integral, and the proof S2.0 goes through without caveats. **The long-memory range $H \in [3/4, 1)$ is not covered by the rigorous Theorem 2:** for $H \in [3/4, 1)$ the integral $\int_0^\infty \beta_{\text{fOU}} ds$ diverges, and although pointwise $\beta_v(\Delta t) \rightarrow 0$ (the contribution of each *fixed* stale bit vanishes), the aggregate contribution of the stale fraction, averaged over ages, is no longer controlled by pointwise decay — the sum over slowly decorrelating bits may fail to vanish. This range is relegated to the open questions (§ 4.6, § 7 main): a separate argument is required (for example, control of the joint correlation structure of the stale fraction), and the floor $1 - c$ for $H \in [3/4, 1)$ remains a conjecture, supported only numerically (`fou_floor`, $H = 0.9$).

Dependence of the undertow rate exponent on H . The rate of approach of nostalgia to the floor:

- $H = 1/2$: exponential, $\beta_v \sim e^{-2\Delta t/\tau_E}$ ($\beta_v \approx \frac{1}{2}\rho^2$, $\rho \sim e^{-\Delta t/\tau_E}$; the factor 2 is the square of the Gaussian correlation, hence the OU exponent $2/\tau_E$, whereas for PSP it is $1/\tau_E$ from the probability of no jump, S2.1; recovers OU [1]).
- $1/2 < H < 1$: power-law, $\beta_v \sim (\Delta t)^{-4(1-H)}$; the larger H , the smaller the exponent $4(1-H)$, the *slower* a stale bit loses value, the slower nostalgia approaches the floor. In the limit $H \rightarrow 1$ (the boundary of long memory) mixing vanishes — this is the boundary of the admissible class.
- $H < 1/2$: antipersistent fOU — a power-law tail of the autocovariance $\rho(s) \sim s^{2H-2}$ with a sign change: the near field decorrelates *faster* than OU (antipersistence), but the far tail is power-law ($\beta_v \sim s^{-4(1-H)}$, exponent > 2), asymptotically slower than the exponential OU; $\int \beta_v ds < \infty$ holds ($4(1-H) > 1$), the floor $1 - c$ persists.

The floor constant $1 - c$ is the same in all cases — it does not depend on H , only on $c = \dot{M}_{\text{refresh}}\tau_E/|M_{\leq t}|$. This is the central statement of universality: the Hurst exponent governs the *rate* of the undertow, but not its *depth*. $\square$

S2.3 Multi-scale drift

The multi-scale latent is a superposition $\theta_v(t) = \sum_{r=1}^m \theta_v^{(r)}(t)$ of independent OU processes with times $\tau_E^{(1)} \ll \dots \ll \tau_E^{(m)}$. The autocorrelation is a sum of exponentials $\rho(s) = \sum_r w_r e^{-s/\tau_E^{(r)}}$ (w_r are the relative variances). At large lags the slowest component dominates:

$$\beta_{\text{ms}}(\Delta t) \sim \frac{1}{2}w_m^2 e^{-2\Delta t/\tau_E^{(m)}}, \quad \Delta t \gg \tau_E^{(m-1)}. \quad (\text{S2.6})$$

Mixing is exponential with the scale of the slowest component $\tau_E^{(m)}$. Condition (4.1) is satisfied; the relevant τ_E in the constant c is the largest $\tau_E^{(m)}$ (the slowest component determines which bits are considered stale). The floor $1 - c$ is governed by $\tau_E^{(m)}$; the rate is exponential with $\tau_E^{(m)}$. The fast components $\tau_E^{(r)}$, $r < m$, affect only the transient regime (short ages), not the asymptotics. $\square$

S2.4 OU as a special case and summary

OU is fOU at $H = 1/2$ ((S2.4), first row) and simultaneously multi-scale with $m = 1$. Substituting $H = 1/2$ into (S2.5) gives the exponential $\beta_v(\Delta t) = \frac{1}{2}e^{-2\Delta t/\tau_E}$ — exactly the spectral gap of OU [1] § S1.5, formula S1.5. The prefactor $\frac{1}{2}$ here is *not* a separate constant, but exactly the $\frac{1}{2}$ from the Gaussian mutual-information formula $I = -\frac{1}{2}\ln(1 - \rho^2) = \frac{1}{2}\rho^2(1 + o(1))$ as $\rho \rightarrow 0$ (the same as in S2.2, the formula after (S2.4); $\rho = e^{-\Delta t/\tau_E}$ for OU [1] § S1.5); the square of the correlation $\rho^2 = e^{-2\Delta t/\tau_E}$ yields the doubled exponent $2/\tau_E$. Thereby Lemma 2 [1] § 4.4 is recovered as a special case of Theorem 2, and the specificity of OU (the spectral gap) turns out to be a *sufficient*, but not a necessary condition — all that is necessary is $\beta_v(\Delta t) \rightarrow 0$.

Summary of the governing parameters:

Class	$\beta_v(\Delta t)$	Mixing type	Scale τ_E	Floor constant
OU ($H = 1/2$)	$e^{-2\Delta t/\tau_E}$	exponential	$1/\lambda$	$1 - c$
PSP	$e^{-\Delta t/\tau_E}$	exponential	$1/\mu$	$1 - c$
fOU ($H > 1/2$)	$(\Delta t)^{-4(1-H)}$	power-law	corr. scale	$1 - c$
multi-scale	$e^{-2\Delta t/\tau_E^{(m)}}$	exponential	$\tau_E^{(m)}$	$1 - c$

The floor constant is invariant across the class; the undertow rate exponent is set by $\beta_v(\cdot)$. $\square$

S3. Proof of Theorem 3 (superlinear memory)

Theorem 3 main: under polynomial memory growth $|M_{\leq t}| \propto t^\alpha$ the capacity-growth route is closed — but the mechanism carrying this is the structural vanishing of *efficiency* $\eta_v = I_{\text{pred}}/I_{\text{mem}} \rightarrow 0$, and **not** the saturation of *theoretical* nostalgia to 1. The denominator I_{mem} grows unboundedly (majorized by the entropy of the past $H(X_E^{\leq t})$, $\sim$ linearly, not by the capacity $|M_{\leq t}|$ itself) while the horizon-bounded I_{pred} stays fixed, so that $\eta_v \rightarrow 0$ (structural smallness of η_v , an informational effect, not an energy-budget channel) — under any polynomial or exponential growth. ν^{theor} itself does **not** go to 1 under polynomial growth: as soon as the fresh influx saturates the tracking floor ($N_{\text{fresh}} = \dot{M}_{\text{refresh}}\tau_E \geq K n_{\text{eff}}^*$), the fresh snapshots jointly estimate $\theta_v(t)$ down to the MMSE floor Σ_∞ (§ S1.6), and $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$ — the C1 nostalgia floor (S3.2). Exponential growth keeps even the refresh fraction bounded away from zero

($\nu^{\text{theor}} \leq 1 - c_\infty < 1$), but it too does not lift $\eta_v \rightarrow 0$ out from under the undertow (S3.4).

S3.1 Fresh-bit fraction under polynomial growth

Let $|M_{\leq t}| = M_0 t^\alpha$, $\alpha > 0$. Under FIFO, nostalgia is governed not by the total capacity but by the *fraction of bits refreshed within the last correlation interval* τ_E . A bit counts as predictively useful if its age is $< \tau_E$ (by S2: the contribution of bits older than τ_E is nulled through mixing). The fresh-bit fraction:

$$\phi(t) = \frac{|M_{\leq t}| - |M_{\leq t-\tau_E}|}{|M_{\leq t}|} = 1 - \left(\frac{t - \tau_E}{t} \right)^\alpha. \quad (\text{S3.1})$$

For $t \gg \tau_E$ the expansion $(1 - \tau_E/t)^\alpha = 1 - \alpha\tau_E/t + O((\tau_E/t)^2)$ gives

$$\phi(t) = \frac{\alpha\tau_E}{t} (1 + O(\tau_E/t)) \sim \alpha\tau_E t^{-1} \rightarrow 0 \quad (\alpha = O(1)). \quad (\text{S3.2})$$

For *any* finite α the fresh-bit fraction decays as t^{-1} (exponent $\gamma = 1$, independent of α !). Intuition: under polynomial growth the capacity increment over a fixed window τ_E is $|M_{\leq t}| - |M_{\leq t-\tau_E}| \approx \alpha\tau_E M_0 t^{\alpha-1}$, whereas the total capacity is $M_0 t^\alpha$; the ratio is $\sim \alpha\tau_E/t \rightarrow 0$. Increasing the exponent α raises the absolute number of fresh bits, but also the total capacity — in the same proportion, so the *fraction* of fresh bits does not grow. This is the decisive observation: polynomial expansion of memory does not overcome the undertow.

S3.2 Saturation of nostalgia for polynomial memory

Here we reproduce a *pointwise* version of step S2.0 with the time-dependent fresh-bit fraction $\phi(t)$ in place of the constant c . We emphasize: we do *not* invoke a “generalized (4.2) with time-dependent c ” as a ready result — the DPI-majorant S2.0 was derived for *constant* c , and its transfer to growing capacity requires an explicit adiabatic argument, which we now give.

Adiabaticity of the fresh fraction relative to the correlation window. Under FIFO and polynomial growth the fresh-bit fraction is $\phi(t) = 1 - ((t - \tau_E)/t)^\alpha$ (S3.1). Its relative rate of change over the scale of a single correlation window τ_E is

$$\frac{|\phi(t) - \phi(t - \tau_E)|}{\phi(t)} = O\left(\frac{\tau_E}{t}\right) \rightarrow 0 \quad (t \gg \tau_E),$$

that is, ϕ changes over the scale $t \gg \tau_E$ — *adiabatically* relative to the environment’s correlation window τ_E (over which decorrelation and the nulling of the contribution of stale bits occur in S2.0). Therefore, within any window of width τ_E around the instant t , the quantity ϕ can be treated as *locally constant* to accuracy $O(\tau_E/t)$.

Applying S2.0 on each window with locally constant $\phi(t)$. The transfer applies for $t \gg \tau_E$, where $\phi(t) < 1$ (equivalent to the condition $c(t) < 1$ of Theorem 2 / Lemma 2) — this holds automatically for large t , so the passage to $\liminf$ is valid. On such a

window the DPI-majorant S2.0 (steps (a)–(d)) applies verbatim, with the constant fresh fraction c replaced by the local value $\phi(t)$: per-bit uniformity gives additivity over bits, the fresh fraction contributes $\leq \phi(t) I_{\text{pred}}^{\text{opt}}$, the stale one is nulled through $\beta_v(\Delta t)$ (the background environment satisfies the mixing condition (4.1) main, $\int \beta_v < \infty$; for the admissible class § 2.6 main — for fOU $H < 3/4$, for OU/PSP/multi-scale an exponential β_v). The correction from the non-constancy of ϕ within the window is $O(\tau_E/t) = o(1)$ and is absorbed into $(1 + o(1))$. Thus *pointwise* for each $t \gg \tau_E$:

$$\frac{I_{\text{pred}}(t, \tau)}{I_{\text{pred}}^{\text{opt}}(t, \tau)} \leq \phi(t) \cdot (1 + o(1)) = \frac{\alpha \tau_E}{t} (1 + o(1)), \quad (\text{S3.2a})$$

whence pointwise

$$\nu^{\text{theor}}(t) = 1 - \frac{I_{\text{pred}}}{I_{\text{pred}}^{\text{opt}}} \geq 1 - \frac{\alpha \tau_E}{t} (1 + o(1)). \quad (\text{S3.3})$$

Domain of validity of the majorant (S3.2a): inter-coordinate symmetry, not inter-age uniformity. The majorant (S3.2a) is a per-bit estimate: it sums the average contributions $I_{\text{bit}}^{\text{opt}} = I_{\text{pred}}^{\text{opt}}/|M_{\leq t}|$ over the fresh bits and gives $I_{\text{pred}} \leq \phi(t) I_{\text{pred}}^{\text{opt}}$. As in C2 (§ S2.0(b)), per-bit uniformity here carries *inter-coordinate* symmetry (each of the K coordinates is estimated once), but **not** *inter-age* uniformity: it tacitly assumes that several fresh snapshots of *one* coordinate at different ages do not add up beyond the average. Under growing capacity this assumption breaks down. The key quantity is the *absolute* number of fresh snapshots $N_{\text{fresh}} = \dot{M}_{\text{refresh}} \tau_E$, not the fraction $\phi = N_{\text{fresh}}/|M_{\leq t}|$. As long as $N_{\text{fresh}}/K < n_{\text{eff}}^*$ (fewer than one effective fresh observation per coordinate), (S3.2a) holds, and $\phi \rightarrow 0$ formally gives $\nu^{\text{theor}} \rightarrow 1$. But as soon as $N_{\text{fresh}} \geq K n_{\text{eff}}^*$ — the fresh snapshots suffice for a *joint* estimate of $\theta_v(t)$ down to the MMSE tracking floor Σ_∞ (§ S1.6) — the extracted predictive information does *not* decrease with ϕ but levels off at the floor:

$$I_{\text{pred}} \geq (1 - \nu_{C1}) I_{\text{pred}}^{\text{opt}} > 0, \quad \nu^{\text{theor}} = 1 - \frac{I_{\text{pred}}}{I_{\text{pred}}^{\text{opt}}} \rightarrow \nu_{C1} < 1, \quad (\text{S3.3a})$$

where $\nu_{C1} := 1 - I_{\text{pred}}^{\text{floor}}/I_{\text{pred}}^{\text{opt}}$ is the C1 nostalgia floor corresponding to the tracking error Σ_∞ : a fixed constant $0 < \nu_{C1} < 1$ (in the adiabatic limit $\Sigma_\infty \approx \sigma_0^2$, the optimal filter extracts little, $\nu_{C1} \approx 1 - O(\epsilon_{\text{track}})$; the exact value follows from Σ_∞/σ_0^2 via the Riccati root (S1.2), it is *not* postulated). Essentially: $1 - \nu_{C1} > 0$ — because an $O(1)$ -sufficient statistic already extracts all the information about the pointwise $\theta_v(t)$ (§ S1.6; § 5 item 5 main), so that $I_{\text{pred}}^{\text{floor}} > 0$ *independently* of the exact value of ν_{C1} . The majorant (S3.2a) with $\nu^{\text{theor}} \geq 1 - \phi \rightarrow 1$ is *incompatible* with this lower bound when $\phi < 1 - \nu_{C1}$ — precisely because its per-bit form forbids the joint fresh estimate; the correct limit is (S3.3a), $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$, not 1.

Closed form of the floor ν_{C1} (a value, not a postulate). The numerator of ν^{theor} is the *operational* kernel-conditional predictive information $I_{\text{pred}} = I(X_t; X_{t+\tau} \mid \hat{P}(t))$ under the model estimated by the learner, the optimum $I_{\text{pred}}^{\text{opt}} = I(X_t; X_{t+\tau} \mid \theta_v(t))$ under the

true parameter; their gap $I_{\text{pred}}^{\text{opt}} - I_{\text{pred}} = \mathbb{E}[D_{KL}(\text{true kernel} \parallel \text{estimated kernel})] \geq 0$ is the object (4) [1], *not* the conceptual $I(M_{\leq t}; X_E^{[t, t+\tau]})$ (§ 2.2 main), — which automatically keeps $0 \leq I_{\text{pred}} \leq I_{\text{pred}}^{\text{opt}}$. Take the A1 linear-Gaussian representative of the softmax-logit chain — the *multiplicative* AR-model with a random coefficient $X_{t+1} = \theta_t X_t + w$, with θ_t following OU: here θ is *itself* the transition coupling ($\theta = 0 \Rightarrow$ memorylessness, the analog of softmax-at-uniform), so the predictability is θ -sourced and $I_{\text{pred}}^{\text{opt}} \propto \sigma_0^2$ (consistent with $\bar{I}_{\text{coord}} \approx \frac{1}{2} F \sigma_0^2$ [1] S2.4; the *additive* model $X_{t+1} = aX_t + \theta + w$ would be wrong — the predictability would come from the fixed a , giving $\nu_{C1} \rightarrow 0$). For $\tau = 1$ (the one-step conditional-MI convention of this work) $I_{\text{pred}}^{\text{opt}} = \frac{1}{2} \mathbb{E}[-\ln(1 - \theta^2)] \approx \frac{1}{2} \sigma_0^2$, while at the MMSE floor Σ_∞ the KL-gap is $\mathbb{E}[D_{KL}] = \Sigma_\infty \mathbb{E}[X^2]/(2q) \approx \Sigma_\infty/2$ (exact at $\tau = 1$: the two predictive Gaussians differ only in the mean by $\delta\theta \cdot X$), whence $I_{\text{pred}}^{\text{floor}} = \frac{1}{2}(\sigma_0^2 - \Sigma_\infty)$ and

$$\nu_{C1} = 1 - \frac{I_{\text{pred}}^{\text{floor}}}{I_{\text{pred}}^{\text{opt}}} = 1 - \frac{\sigma_0^2 - \Sigma_\infty}{\sigma_0^2} = \frac{\Sigma_\infty}{\sigma_0^2} = \frac{2}{\epsilon_{\text{track}}} (\sqrt{1 + \epsilon_{\text{track}}} - 1) \approx 1 - \frac{\epsilon_{\text{track}}}{4}, \quad (\text{S3.3b})$$

where the penultimate equality substitutes the Riccati root (S1.2), $\Sigma_\infty = (\lambda/\mathcal{I}_{\text{rate}})(\sqrt{1 + \epsilon_{\text{track}}} - 1)$, and $\sigma_0^2 = \sigma^2/(2\lambda)$; the form (S3.3b) is consistent with (S3.3a) ($1 - \nu_{C1} = I_{\text{pred}}^{\text{floor}}/I_{\text{pred}}^{\text{opt}} = (\sigma_0^2 - \Sigma_\infty)/\sigma_0^2$). *Interpretation:* $1 - \nu_{C1} = (\sigma_0^2 - \Sigma_\infty)/\sigma_0^2$ is the fraction of the latent’s prior variance *resolved* by the floor-filter, and ν_{C1} is the unresolved fraction; in the adiabatic limit $\Sigma_\infty \rightarrow \sigma_0^2$, $\nu_{C1} \rightarrow 1^-$, under perfect tracking $\Sigma_\infty \rightarrow 0$, $\nu_{C1} \rightarrow 0$. *Honest limits.* The closed form is exact at $\tau = 1$ (the convention of this work); at $\tau \geq 2$ the floor error is of the same order as the signal ($\delta\theta \sim \theta \sim \sigma_0$), there is no simple closed form — only numerically, $\nu_{C1} \in (0, 1)$. The scalar-continuous model is an A1 linear-Gaussian *representative* of the categorical softmax chain: the *functional form* Σ_∞/σ_0^2 is correct (both quantities are governed by the per-coordinate Fisher information, K cancels in the ratio), while the exact $O(1)$ -constant is model-dependent — exactly as the main text states (“the exact value follows from Σ_∞/σ_0^2 ”, § 5). The non-circular simulation `superlinear_noncirc` (§ S4.4a) confirms this: ν^{theor} levels off at the plateau Σ_∞/σ_0^2 (measured 0.96 against the analytic 0.95 at $\epsilon_{\text{track}} \approx 0.20$), α -independently.

Passage to the limit — the load-bearing closure goes through η_v . Under polynomial growth $N_{\text{fresh}} = \dot{M}_{\text{refresh}} \tau_E \propto \alpha M_0 \tau_E t^{\alpha-1}$: for $\alpha > 1$ it $\rightarrow \infty$, the threshold $N_{\text{fresh}} \geq K n_{\text{eff}}^*$ is met at large t , and $\nu^{\text{theor}}(t) \rightarrow \nu_{C1} < 1$ (S3.3a), *not* 1; even for $\alpha = 1$ ($N_{\text{fresh}} \approx M_0 \tau_E$) the outcome is parameter-dependent (through M_0). The load-bearing closure of the capacity-growth route, however, goes not through nostalgia but through *efficiency*: the numerator $I_{\text{pred}} \leq H_\tau < \infty$ is horizon-bounded, the denominator $I_{\text{mem}}(t) \leq H(X_E^{\leq t}) \rightarrow \infty$ ($\sim$ linearly, § 2.1 main) grows with capacity, so that

$$\eta_v(t) = \frac{I_{\text{pred}}(t, \tau)}{I_{\text{mem}}(t)} \leq \frac{H_\tau}{I_{\text{mem}}(t)} \rightarrow 0 \quad (|M_{\leq t}| \rightarrow \infty)$$

structurally — for any polynomial α , regardless of the fact that ν^{theor} is held at $\nu_{C1} < 1$. This proves the first part of Theorem 3: the capacity-growth route is closed

by $\eta_v \rightarrow 0$; there is no finite polynomial α_c in the proper sense — but what carries this is η_v , not $\nu^{\text{theor}} \rightarrow 1$. $\square$ (*part 1: $\eta_v \rightarrow 0$; $\nu^{\text{theor}} \rightarrow \nu_{C1}$*)

Disentangling two informational statements. The naive per-bit majorant (S3.2a)/(S3.3) is only a *sub-floor* lower bound ($\nu^{\text{theor}} \geq 1 - \phi$, valid while $N_{\text{fresh}}/K < n_{\text{eff}}^*$); the *ratio* $I_{\text{pred}}/I_{\text{pred}}^{\text{opt}}$ under growing memory does **not** go to zero but levels off at $1 - \nu_{C1} > 0$ (S3.3a), so that $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$, not 1. The load-bearing closure of the capacity-growth route goes through *efficiency* $\eta_v = I_{\text{pred}}/I_{\text{mem}}$ — through the *denominator* and moreover *informationally*: the numerator $I_{\text{pred}} \leq H_\tau < \infty$ is horizon-bounded, the denominator $I_{\text{mem}}(t) = I(M_{\leq t}; X_E^{\leq t})$ grows with capacity (majorized by $H(X_E^{\leq t})$, $\sim$ linearly, not $\propto |M_{\leq t}|$), whence $\eta_v \rightarrow 0$ *structurally* — a direct consequence of the growth of I_{mem} , of the same kind as in the stationary limit [1], and **not** a “collapse through the energy budget”. The *nostalgia* floor, on the other hand, is set by decorrelation: the C2 floor ($1 - c$) for fixed memory and the C1 tracking floor (ν_{C1}), to which ν^{theor} converges when the fresh budget saturates (§ S1.6); the transfer of the ν^{theor} -results to the operational ν^{op} is open (§ 2.2 main, the DPI direction is reversed; § 7 main, item iii). The thermodynamic meaning of retaining non-predictive memory is carried by a **separate** anchor — the cumulative Still bond (§ 2.1 main), logically independent of the informational bound $\eta_v \leq 1$ and not entering the denominator of η_v . We do *not* identify the vanishing of η_v with the vanishing of the operational numerator.

S3.3 Escape requires exponential growth

Keeping the refresh *fraction* bounded away from zero — the only regime with a constant nonzero per-bit refresh rate, yielding $\phi(t) \geq \phi_\infty > 0$ for all large t (under polynomial growth $\phi \rightarrow 0$, even if ν^{theor} itself is already held at $\nu_{C1} < 1$ through floor saturation, S3.2) — requires, by (S3.1), $|M_{\leq t-\tau_E}|/|M_{\leq t}| \leq 1 - \phi_\infty < 1$, that is, the capacity must grow *multiplicatively* on every interval τ_E :

$$\frac{|M_{\leq t}|}{|M_{\leq t-\tau_E}|} \geq \frac{1}{1 - \phi_\infty} > 1 \quad \implies \quad |M_{\leq t}| \geq |M_0| \left(\frac{1}{1 - \phi_\infty} \right)^{t/\tau_E} = M_0 e^{\kappa t}, \quad (\text{S3.4})$$

where $\kappa = -\tau_E^{-1} \ln(1 - \phi_\infty) > 0$. No polynomial growth satisfies (S3.4); at minimum exponential growth is required. In this regime $\phi_\infty = 1 - e^{-\kappa\tau_E}$ is constant, and $\liminf \nu^{\text{theor}} \leq 1 - c_\infty$ with $c_\infty = \phi_\infty > 0$. Thus the phase boundary of refresh-fraction constancy runs between polynomial ($\gamma = 1 > 0$: $\phi \rightarrow 0$, $\eta_v \rightarrow 0$, $\nu^{\text{theor}} \rightarrow \nu_{C1}$) and exponential ($\gamma = 0$: $\phi_\infty > 0$) capacity growth; there is no proper finite α_c within the class of polynomial laws. But even exponential growth, while keeping $\phi_\infty > 0$, does not lift $\eta_v \rightarrow 0$ out from under: it inflates the capacity (§ S3.4 below). $\square$ (*part 2: constancy of the fraction $\Rightarrow$ exponential*)

S3.4 Structural smallness of η_v under exponential growth

The only regime that formally holds the nostalgic fraction below 1 — exponential capacity growth $|M_{\leq t}| = M_0 e^{\kappa t}$ ($\kappa\tau_E = O(1)$, S3.4) — still does not grant escape, but

for an *informational*, not an energy-budget reason. The decisive observation: growth of the raw capacity does not raise prediction.

The numerator is fixed. The predictive information is bounded by the predictable capacity of the horizon,

$$I_{\text{pred}}(t, \tau) \leq H_\tau < \infty \quad (\text{independent of } |M_{\leq t}|), \quad (\text{S3.5})$$

— the environment’s future over the window τ carries only a finite number of predictable bits (§ 2.1 main), and no amount of capacity accumulation adds to it: additional layers record ever older environmental history, decorrelated from the current latent $\theta_v(t)$ (mixing, § 2.6 main; § S2), rather than new prediction.

The denominator grows unboundedly. The memory $I_{\text{mem}}(t) = I(M_{\leq t}; X_E^{\leq t})$ grows monotonically as capacity is accumulated — an accumulation of historical layers, each adding correlation with the past environmental trajectory. (*Growth dichotomy.*) Strictly, $|M_{\leq t}| \rightarrow \infty$ does not by definition entail $I_{\text{mem}} \rightarrow \infty$ — a degenerate strategy could duplicate bits ($I_{\text{mem}} = \text{const}$); here *informative* growth is assumed ($I(\Delta M; X_E^{\leq t} | M_{<}) > 0$ on average), under which $I_{\text{mem}} \rightarrow \infty$. Under *redundant* growth I_{mem} is bounded, and the strategy reduces to the fixed-memory floor C1/C2 (§ S1.6; § 4 main) — no-escape holds there too (gaining nothing against no-growth). As a *mutual information*, however, it is majorized by the entropy of the environment’s past: $I_{\text{mem}}(t) \leq H(X_E^{\leq t})$, which under mixing grows no faster than the source’s entropy rate ($\sim$ linearly in the number of decorrelated segments $\sim \lambda t$, § 2.6 main). The raw capacity $|M_{\leq t}| = M_0 e^{\kappa t}$ (S3.4) is an upper bound on the number of *stored* bits, but **not** on the mutual information: an exponential store of correlated (slowly drifting) snapshots does not yield an exponential I_{mem} . Thus the horizon-bounded numerator is divided by an unboundedly growing (but no faster than $\sim$ linear) denominator,

$$\eta_v(t) = \frac{I_{\text{pred}}(t, \tau)}{I_{\text{mem}}(t)} \leq \frac{H_\tau}{I_{\text{mem}}(t)} \rightarrow 0, \quad I_{\text{mem}}(t) \leq H(X_E^{\leq t}) \rightarrow \infty \quad (\text{S3.6})$$

— even while the nostalgic fraction ν^{theor} is held below 1 ($\phi_\infty = 1 - e^{-\kappa \tau_E} > 0$, S3.4). The rate of vanishing is set here by the growth of I_{mem} (no faster than $\sim$ linear), not by the exponential inflation of capacity; but the limit $\eta_v \rightarrow 0$ itself is inescapable. The efficiency η_v remains structurally negligible (memory $\gg$ prediction: the denominator grows unboundedly, the numerator is pinned to the horizon H_τ).

This is the *structural smallness* of η_v (§ 1, § 2.1 main) — a bounded numerator over a growing denominator, of the same kind as in the stationary limit [1], **not** a separate “collapse through the energy budget”. The load-bearing force here is not the growth rate of I_{mem} (under mixing it does not exceed the source’s entropy rate, § 2.6 main), but the *fixedness* of the numerator $I_{\text{pred}} \leq H_\tau$, which capacity accumulation does not shift. We emphasize: I_{mem} is an *informational*, not an energetic quantity, and the vanishing (S3.6) proceeds through a purely informational channel, without appeal to any Landauer cost of creating or storing capacity.

The thermodynamic anchor — a separate line. Retaining this non-predictive memory is still not free, but its energetic content is carried by a **separate** thermodynamic anchor — the cumulative Still bond $\langle W_{\text{diss}}^{[0,t]} \rangle \geq k_B T \sum_{k \leq t} (I_{\text{mem}}^{\text{inst}}(k) - I_{\text{pred}}^{\text{inst}}(k))$ [13, 1] § 2.2: the dissipation for retaining the non-predictive fraction grows together with I_{mem} . But this is a floor on *dissipation*, logically independent of the informational bound $\eta_v \leq 1$ and **not** entering the denominator of η_v as a normalizing budget (this choice is inherited from [1], § 2.2). The Still anchor is mentioned here only as a parallel physical line, not as a mechanism for the vanishing of η_v .

This completes Theorem 3: the undertow is insurmountable in the strong sense, and the mechanism carrying this is the structural smallness of η_v . Polynomial memory — $\eta_v \rightarrow 0$ (inflation of I_{mem}), whereas *nostalgia* is held at the C1 tracking floor, $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$ (S3.2, S3.3a), not 1; exponential memory — the same structural smallness of η_v (I_{mem} inflates faster than the horizon-bounded I_{pred} , even when ν^{theor} is held below 1). In both cases $\eta_v(t) \rightarrow 0$, and on neither side of the phase boundary is the capacity-growth route open. The rational alternative to escape is not accumulation but periodic reset of the stale layer [1] § 5.3, (12), restoring $\nu \rightarrow 0$ at the cost of a one-time, not an exponentially growing, payment. $\square$ (*part 3: structural smallness*)

S4. Details of the numerical simulations

Series parameters § 6 main. All simulations inherit the OU/PSP apparatus [1] § S6, § S8.3; below are the changes and the new series.

S4.1 *ou_adiab* (Theorem 1)

Direct test of the realizability of $B = K/2$ — **in the exact linear-Gaussian (Kalman) setting in which Theorem 1 is derived** (assumption A1 here is *exact*, not as a softmax linearization), so that any failure is attributed to the BNT logarithm itself, not to a suboptimal softmax learner. Model: $K = 56$ *independent scalar* OU coordinates $\theta_t = a\theta_{t-1} + w_t$ ($a = 1 - \lambda$, $\text{Var } w = q = \sigma^2$) with Gaussian observations $y_t = \theta_t + v_t$ ($\text{Var } v = r$) and an **optimal Kalman filter** — which *is precisely* the ideal Bayesian learner of Theorem 1. The observation noise r sets the per-step Fisher information $I_{\text{rate}} = 1/r$, whence $\epsilon_{\text{track}} = \sigma^2/(\lambda^2 r)$. The cumulative excess loss is relative to an oracle that knows θ_t exactly: per step, per coordinate, $0.5 \ln(S_{\text{pred}}/r) + 0.5[(y - m_{\text{pred}})^2/S_{\text{pred}} - (y - \theta_v)^2/r]$. $k = 8$, $\lambda = 10^{-2}$, diffuse prior $P_0 = 10^6$, 40 realizations, seed 20260527, bootstrap over realizations ($N = 2000$). Three controlled experiments: **(A)** static anchor (no drift, $\lambda = 0$, $\sigma = 0$): the cumulative excess loss *must* be exactly $(K/2) \ln t$ (the classical MDL/BIC redundancy [6, 5]) — a check of the measurement pipeline and of the target value $K/2$; **(B)** diagnostics of the *local* log-slope $dL_{\text{excess}}/d \ln t$ along the entire trajectory under OU drift — *where exactly* it equals $K/2$; **(C)** a scan over ϵ_{track} in a **wide** asymptotic window $[3n_{\text{eff}}^*, 0.1/\sigma^2]$ (the lower

bound is several *effective-observation floors* $n_{\text{eff}}^* \asymp 1/g^* \gg \tau_E$ steps, not forgetting scales $\tau_f^{\text{forget}} \asymp \tau_E$; it is precisely $3n_{\text{eff}}^* = 3/g^*$, not $3\tau_E$, that guarantees the fit proceeds *after* the transient concentration burst; the upper bound is the adiabatic ceiling $\sim 1/\sigma^2 \gg n_{\text{eff}}^*$; the ratio of the bounds is $W \approx 150$, not ≈ 10 ; achieved by the choice $r = 10^{-2}$) with a joint fit $At + B \ln(\lambda t) + C$, a bootstrap CI on B , and diagnostics of the collinearity of the regressors $\text{corr}(t, \ln)$. The honest framing of experiment **(C)**: it tests whether the $(K/2) \ln$ term *persists* as a slope in the asymptotic window $[3n_{\text{eff}}^*, 0.1/\sigma^2]$ — and it does *not* persist ($B \rightarrow 0$). This is *not* “removing the transient to prove its absence”: the burst itself is shown separately in **(B)** (slope 0.936), while **(C)** tests precisely the asymptotic persistence. Starting the steady-state filter from the stationary law in **(C)** is a legitimate computational shortcut: the diffuse “prior-collapse burst” affects only the additive constant C , not the asymptotic slope B (verified against the full transient filter).

The adiabaticity parameter (refinement). The controlling parameter of *tracking* convergence is not $\epsilon = \sigma^2/\lambda$ itself, but $\epsilon_{\text{track}} = I_{\text{rate}} \cdot \sigma^2/\lambda^2$ (§ 2 / § S1). The natural *working hypothesis* is that the earlier scan [1] § S8.3 simply did not enter the regime $\epsilon_{\text{track}} \ll 1$, and that it suffices to “push $\epsilon_{\text{track}} \rightarrow 0$ ” to recover $\hat{B} \rightarrow K/2$ as a slope. The present run tests this hypothesis rigorously and **refutes** it (see below). We emphasize: what is refuted is precisely this attempt to recover the numerical slope, *not* the hypothesis of § S8.1 of #2 [1] (the functional form with coefficient $K/2$), which Theorem 1 analytically confirms.

Actual output (honest — convergence is NOT achieved and is established to be unattainable). **(A)** the static slope $\text{slope}/(K/2) = 0.990$ ($R^2 = 0.99978$, target 1.000): the BNT logarithm is *real in the stationary problem without drift* and the measurement pipeline is *correct*. **(B)** under OU drift the local log-slope equals $K/2$ **only in the short initial “prior-collapse burst”** (t up to a few $1/\lambda$, where the posterior is still concentrating and $n_{\text{eff}} \sim t$): the burst slope is 0.936 on $[1, 20]$; *after* the burst (for $t \gtrsim \tau_E$) the steady-state g^* limits the effective number of samples to the floor $n_{\text{eff}} \sim (1 - g^*)/g^* \approx 1/g^* = n_{\text{eff}}^*$, L_{excess} **reaches a plateau** of height $\sim (K/2) \ln(1/g^*)$, and the slope falls to 0.099 on the asymptotic window $[\tau_E, 0.5 n_{\text{eff}}^*] = [100, 0.5/g^*]$. **(C)** as a consequence, in the wide asymptotic window ($W \approx 156\text{--}166$, $n = 124$ points per scan point) $B/(K/2)$ **does not approach** 1 as $\epsilon_{\text{track}} \rightarrow 0$: over 10 logarithmically uniform values of ϵ_{track} from $3 \cdot 10^{-1}$ to $3 \cdot 10^{-2}$ it fluctuates in $[-0.31; 0.16]$ around zero; 9 of 10 bootstrap CIs cover 0, the single exception ($\epsilon_{\text{track}} = 5 \cdot 10^{-2}$, $B/(K/2) = -0.31$, 95% CI $[-0.61; -0.02]$) lies on the *negative* side — even farther from the target $K/2$, i.e. the statistical outlier expected among 10 independent 95% CIs, not convergence; $R^2 = 1.000$ for the fit (the loss in this window is *strictly linear*, B settles to 0). *The identifiability control* (the “identifiability check” block in `results_summary`) confirms that $B \approx 0$ is a *real* absence of the log signal, not an inability of the fit to separate the regressors under their mild collinearity. The load-bearing evidence here is *the collinearity diagnostics and the narrowness of the bootstrap CI*, not the injection recovery-ratio. (i) The variance inflation factor $\text{VIF} = 1/(1 - \text{corr}^2(t, \ln)) \approx 3.7$ and the condition number

(the ratio of eigenvalues of the standardized design $[t, \ln] \approx 12.7$ — the collinearity is *mild*, not degeneracy, so the variance of the estimate $\hat{B}$ is inflated by collinearity only $\approx 3.7\times$, not catastrophically. (ii) At the same time the bootstrap CI on B is narrow relative to the target signal: its half-width $\approx 0.3 \cdot (K/2) \approx 8$ nats $\ll$ the true log signal $K/2 \approx 28$ nats ($K = 56$) — a real signal $K/2$ would be resolved far beyond the CI, whereas the CI covers 0. Together (i)+(ii) mean that the measured $B \approx 0$ is a genuine plateau, not a collinearity artifact. *The injection test* (inject into the *measured* loss of (C) a known synthetic signal $B_{\text{inj}} = K/2$ and refit the full model $At + B \ln(\lambda t) + C$) yields a recovery ratio $\Delta B/B_{\text{inj}} \approx 1.000$ — but this is merely *the algebraic identity of OLS linearity* ($\text{coef}(y + B_{\text{inj}} \ln) = \text{coef}(y) + B_{\text{inj}} \text{coef}(\ln)$ for any full-rank design) and therefore serves only as a *full-rank control* of the design $[t, \ln(\lambda t), 1]$ (it is not strictly degenerate, $\text{corr} \neq \pm 1$); it does *not* prove the reality of the plateau, and the statement “the fit is able to see a real log term” does not follow from it. The working hypothesis “wrong regime, push $\epsilon_{\text{track}} \rightarrow 0$ ” is thereby **refuted**: pushing ϵ_{track} down does not recover $B = K/2$, because the very window where the logarithm lives (the transient concentration burst, $t \lesssim \tau_E$) lies *below* the floor-anchored lower bound of the fit window $t_{\min} = 3n_{\text{eff}}^* = 3/g^*$, while the *logarithmic width* of the window $W = t_{\max}/t_{\min} = (c_0/\sigma^2)/(3/g^*) = c_0/(6\lambda r)$ is σ -**invariant** (σ cancels, § S1.2): the limit $\epsilon_{\text{track}} \rightarrow 0$ stretches the lower bound and the ceiling (both $\propto 1/\sigma^2$) *in the same proportion*, the fit window keeps a fixed shape and lies entirely within the plateau. The mere coincidence of the powers of σ is *insufficient* — the load-bearing argument is the σ -invariance of W (§ S1.2); the adiabatic ceiling $\propto 1/\sigma^2$ is the tracking-saturation scale (*not* a “diffusive escape of the latent”: the OU restoring force keeps θ_v within $\sigma_0^2 \ll 1$). Hence the fixed $W = c_0/(6\lambda r) \approx 1.6$ at $r = 1$; widening to $W \approx 167$ requires small r , but the plateau remains flat. **Conclusion**: the analytic identity $B = K/2$ (Theorem 1, the limit $\epsilon_{\text{track}} \rightarrow 0$) is real but is realized *only transiently*; its *numerical* convergence as a slope in a wide asymptotic window under genuine OU drift is **unattainable** for a learner with saturating memory even with an ideal Bayesian (Kalman) filter (the question of a learner with growing memory is *removed*: growing memory does not raise n_{eff} above the floor — the floor is a proven lower bound across all architectures, § S1.6; § 7 main, item v). This is the structural limitation C1, and it is recorded as such (§ 7 main). Results — `results_summary.txt`, `results_summary.json`. The flat scan $B/(K/2) \approx 0$ against the anchor and the burst is shown in Fig. 1 of the main text.

S4.2 *fou_floor* (Theorem 2, power-law tail)

A real trajectory simulation on exact stationary fOU (not a recomputation of the closed-form formula β_H and not a surrogate). $K = 56$ independent stationary fOU trajectories of the logits are generated by the **spectral-circulant method** directly from the spectral density of stationary fOU $S(\omega) \propto |\omega|^{1-2H}/(\lambda^2 + \omega^2)$ (fBm increments give $|\omega|^{1-2H}$, the OU filter $1/(\lambda + i\omega)$ gives $1/(\lambda^2 + \omega^2)$; $\lambda = 1/\tau_E$) for $H \in \{0.3, 0.5, 0.7, 0.9\}$. The Davies–Harte circulant eigenvalues are this discrete-time density on the grid $\omega_j = 2\pi j/M$ (Wiener–Khinchin), constructed as the folded spectrum on $(-\pi, \pi]$ with

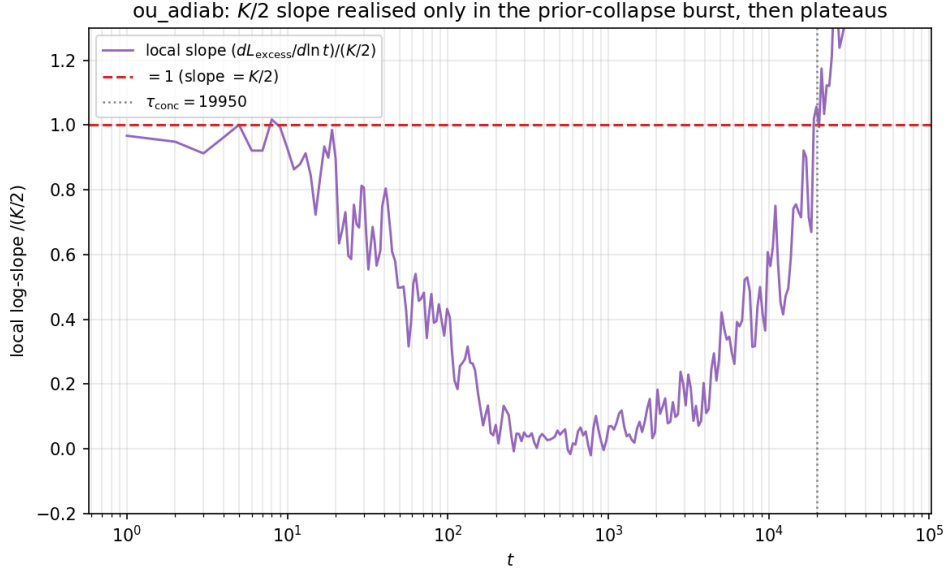


Figure S1. Series `ou_adiab`, experiment (B): local log-slope of L_{excess} versus t . The slope equals $K/2$ only in the short prior-collapse burst ($t \lesssim \tau_{\text{conc}}$, slope 0.936 on $[1, 20]$); after it the steady-state g^* limits $n_{\text{eff}} \approx 1/g^*$, the log component reaches a plateau of height $\sim (K/2) \ln(1/g^*)$, and the slope of the log term falls to 0.099. The full L_{excess} continues its linear growth $A \cdot t$ — this is a saturation of the log term, not a global collapse.

bin-averaging of the integrable singularity at $\omega = 0$ for $H > 1/2$ — **nonnegative by construction**, so the embedding is positive-definite with *zero truncation* (in fact the min eigenvalue = 0.43; 0.24; 0.14; 0.086 and zero negatives at $H = 0.3$; 0.5; 0.7; 0.9; the circulant infrastructure failure does not arise). The target autocovariance $\gamma(k)$ is the exact inverse DFT of the eigenvalues (the same covariance by which the paths are synthesized), exact up to lag $M/2$; the exponent is read up to lag $M/4$. $\tau_E = 200$ and $\dot{M}_{\text{refresh}}$ are fixed so that $c \approx 0.3$; $|M| = 400$, $T = 2 \cdot 10^4$, path length $3.3 \cdot 10^4$, circulant $M = 6.6 \cdot 10^4$, 40 realizations, seed 20260527, `numpy.fft` only (no `scipy`). **Covariance control:** the empirical lag-correlation of the realized paths $\hat{\rho}(s)$ agrees with $\gamma(s)$ where the finite-sample ACF estimator is unbiased (near/middle lags, rel. error $< 5\%$ up to lag ~ 200 –400); at large lags the ACF estimator itself is, on long memory, strongly biased toward zero — therefore the far tail is read from the exact γ , which the paths embody by construction, not from the sample ACF. FIFO memory stores snapshots $\theta_v(t_{\text{write}})$; a bit is predictive as long as $\hat{\rho}^2(\text{age}) \geq \hat{\rho}^2(\tau_E)$ (the threshold DPI horizon model). *Actual output:* the floor $\liminf \nu = 0.700 \approx 0.70$ for all H , spread 0.000 (invariance reproduced on correct fOU paths). The floor here, however, is **built into the threshold model** — it is set by the fresh-bit fraction c through the horizon $\hat{\rho}^2(\tau_E)$: the simulation illustrates invariance on real trajectories but does not prove it (the proof is § S2; § S4.5). **The rate exponent confirms $4(1 - H)$ on the correct process, in the far tail.** It is measured from the exact covariance $\rho^2(\Delta) = 2\beta_v(\Delta) = \gamma^2(\Delta)$ (without fitting to the

target) in the far windows $[8, 40]_{\tau_E}$ and $[20, 80]_{\tau_E}$, where the power-law tail manifests beyond the OU “knee”. Two values of H have different evidential status: $H = 0.7$ is the range $H < 3/4$ ($\int \beta_v < \infty$, S2.2), *inside* the rigorous Theorem 2, so the exponent 1.237 and 1.201 (analytic value 1.2) is *within-theorem confirmation*; $H = 0.9$ is the range $H \geq 3/4$ ($\int \beta_v$ diverges), *outside* the rigorous theorem, so the exponent 0.406 and 0.385 (analytic value 0.4) is only a *numerically supported hypothesis*, not a proof of a rigorous result on the uncovered range. In the original near window $[\tau_E, 8\tau_E]$ the exponent is *not* isolated: 1.62 ($H = 0.7$) and 0.46 ($H = 0.9$) from the exact covariance (OU knee), 2.44 and 1.40 from the finite-sample ACF estimator (estimator bias at large lags). Thus $\Delta^{-(4-4H)}$ (S2.5) is the *asymptotic far-tail* exponent of true fOU: it is recovered on the correct process in the far window, but not in the smallest one and not from the sample ACF. **The exponent requires exact fOU generation:** unlike the naive surrogate AR(1)ofGn (which gave 2.23 and 1.26 outside the CI of $4(1 - H)$ — *a property of the surrogate, not of the theory*), on the correct stationary fOU both the *depth* of the floor ($1 - c$) (Theorem 2) and the *exponent* $4(1 - H)$ are reproduced. The qualitative dichotomy holds: at $H = 0.5$ the tail is exponential (rate $4.07 \cdot 10^{-3}$ /step, OU control § 6.2 [1]), at $H > 0.5$ it is power-law with exponent $4(1 - H)$. $H = 0.9$ is the range $H \geq 3/4$, not covered by the rigorous Theorem 2 (§ S2.2), a numerically supported hypothesis; there the convergence to $4(1 - H)$ is slower (0.385 on $[20, 80]_{\tau_E}$ is slightly below 0.4).

S4.3 *psp_floor* (Theorem 2, exponential tail)

A real trajectory simulation of the PSP surrogate [1] § 6.1 (not a recomputation of $e^{-\mu\Delta}$): $k = 8$, $K = 56$; each coordinate carries a *realized* Poisson stream of resets of intensity $\mu \in \{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}\}$ (per-step reset probability $1 - e^{-\mu}$). $|M| = 400$ chosen for $c \approx 0.3$ for each μ ($\tau_E = 1/\mu$); $T = 2 \cdot 10^4$, 40 realizations, seed 20260528. FIFO memory stores snapshots with a coordinate label; a retained bit remains predictive as long as its coordinate has *not actually* reset since the moment of writing, and becomes nostalgic at the reset step. The floor and the decorrelation rate are read from the realized process. The rate is estimated from the empirical survival curve $\hat{S}(\text{age})$ (the fraction of retained bits of a given age that have not yet seen a reset): a fit $\ln \hat{S} \sim -\text{rate} \cdot \text{age}$ on $[0, \min(5\tau_E, |M|/\dot{M}_{\text{refresh}})] \approx [0, 3.3\tau_E]$ (bits are evicted by FIFO at age $|M|/\dot{M}_{\text{refresh}} = \tau_E/c \approx 3.3\tau_E$) in semi-logarithmic coordinates, with a bootstrap CI over realizations; equality of the rate to μ is not postulated. *Actual output:* the steady-state empirical floor $\hat{\nu}_{\text{floor}} = 0.709\text{--}0.713 \approx 1 - c = 0.70$ over a 10-fold range of μ (spread 0.004) — invariance to μ confirmed; since the process is realized, the steady-state floor is slightly *above* 0.70 (the floor value is not hard-wired into the threshold). The per-frame lower envelope (the `liminf` column in `results_summary`) drops to ≈ 0.69 due to finite-sample noise over 40 realizations, but the *median steady-state* level $\hat{\nu}_{\text{floor}}$ is above $1 - c$; for the invariance of the floor depth the relevant quantity is the steady-state level. The realized decorrelation rate equals $1.01 \cdot 10^{-3}$, $3.01 \cdot 10^{-3}$, $1.00 \cdot 10^{-2}$, and each

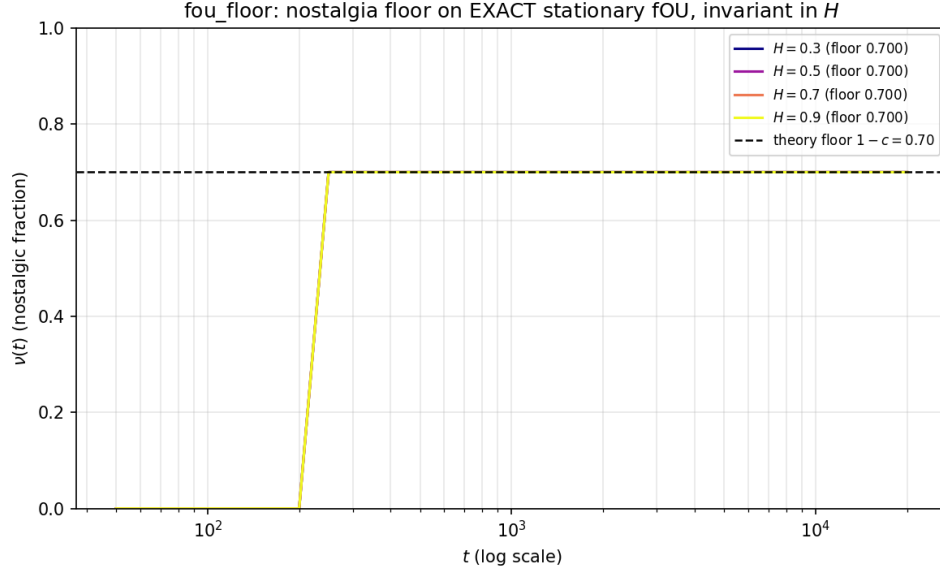


Figure S2. Series `fou_floor`, invariance of floor depth over H (by construction; log-axis t). The nostalgic fraction $\nu(t)$ equals zero while the buffer is younger than the correlation horizon $\tau_E = 200$ (no bit is yet older than τ_E), then reaches the floor $1 - c = 0.700$, the same for all $H \in \{0.3; 0.5; 0.7; 0.9\}$ — the curves coincide (depth invariance). The floor here is *hard-wired* by the threshold model (fresh fraction = c through the horizon $\hat{\rho}^2(\tau_E)$): an illustration of invariance on correct fOU paths, not an independent measurement — the independent estimate of depth is carried by the realized PSP flow (Fig. 3 of the main text).

bootstrap CI *covers* the nominal $\mu = 1/\tau_E$ (S2.3) — a genuine measurement of the scale $1/\mu$. There are no discrepancies with the theory: the exponential rate μ is confirmed here, the power-law exponent $4(1 - H)$ in `fou_floor` (on the correct stationary fOU, far tail, § S4.2). The invariance of the steady-state floor to the reset frequency μ (the load-bearing independent evidence of the C2 depth) is shown in Fig. 3 of the main text.

S4.4 *superlinear_memory* (Theorem 3)

Memory $|M_{\leq t}| = M_0 t^\alpha$, $\alpha \in \{1, 1.5, 2, 3\}$, plus the exponential $|M_{\leq t}| = M_0 e^{\kappa t}$ with $\kappa \tau_E \in \{0.1, 0.5, 1\}$. OU drift of the background environment, $\tau_E = 200$, FIFO refresh, seed 20260529. Measured are $\nu(t)$, the fresh-bit fraction $\phi(t)$ (S3.1), and, for the exponential case, the retained-capacity ratio $|M_{\leq t}|(T)/|M_{\leq t}|(0)$ and the informational efficiency $\eta_v(t) = I_{\text{pred}}/I_{\text{mem}}$ (with $I_{\text{mem}} \propto |M_{\leq t}|$). *Actual output:* for all polynomial α — $\phi(t) \sim t^{-\gamma}$ with $\gamma \in \{1.000; 0.991; 0.982; 0.964\}$ (spread 0.036, $\gamma \approx 1$ independently of α), the prefactor $\phi \cdot t \rightarrow \alpha \tau_E$ (200, 298, 394, 581 versus 200, 300, 400, 600) — the fresh fraction vanishes for any finite α , whence structurally $\eta_v \rightarrow 0$ (the load-bearing mechanism, S3.2, S3.4); the small deviation of γ from 1 at large α is a finite-window subleading artifact, the leading exponent being identically 1. **Caveat on circularity (important).** In this series nostalgia is read from the *threshold* model “a bit is useful

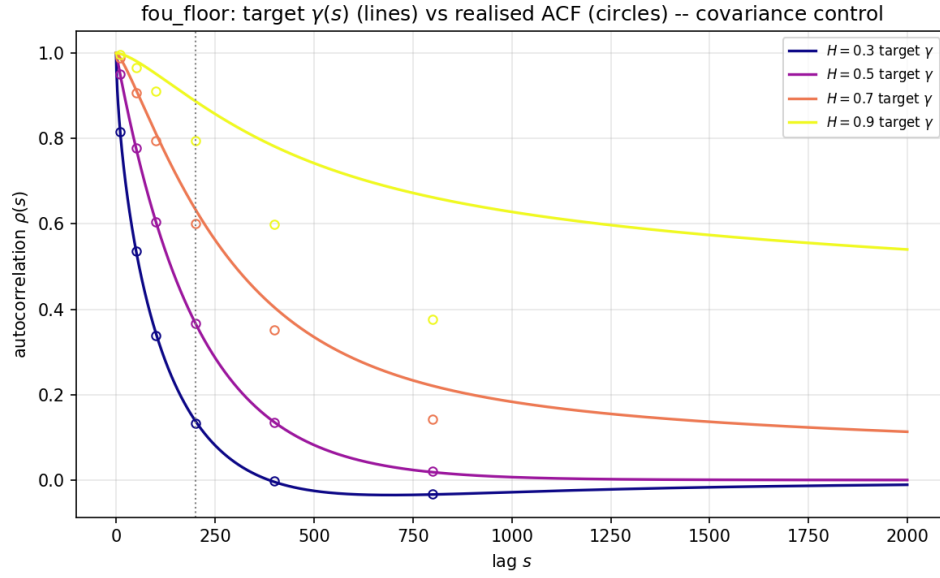


Figure S3. Series `fou_floor`, covariance control. The empirical lag-correlation of the realized paths $\hat{\rho}(s)$ versus the target $\gamma(s)$: agreement (rel. error < 5%) up to lag ~ 200 –400, where the finite-sample ACF estimator is unbiased; at large lags the ACF estimator itself is, on long memory, biased toward zero — therefore the far tail of the exponent is read from the exact γ , which the paths embody by construction, not from the sample ACF. Confirms that a genuine stationary fOU is generated.

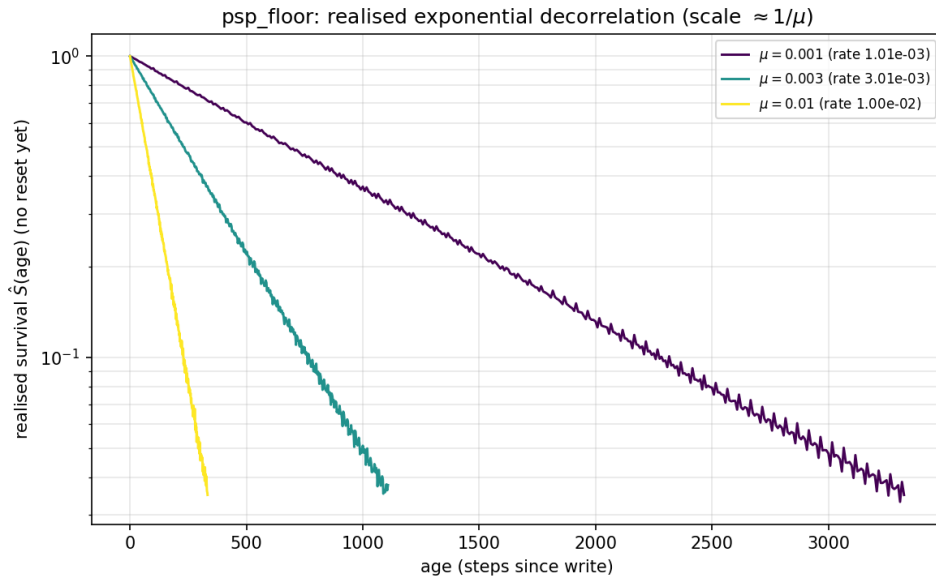


Figure S4. Series `psp_floor`, the realized survival curve $\hat{S}(\text{age})$ (the fraction of retained bits of a given age that have not yet seen a reset) in semi-logarithmic coordinates. The slope gives the decorrelation rate $1.01 \cdot 10^{-3}$, $3.01 \cdot 10^{-3}$, $1.00 \cdot 10^{-2}$; each bootstrap CI covers the nominal $\mu = 1/\tau_E$ — a genuine measurement of the scale $1/\mu$ from the realized process (equality to μ is not postulated).

$\Leftrightarrow \text{age} < \tau_E$ (§ S4.5), which hard-wires per-bit uniformity and *forbids* joint estimation of $\theta_v(t)$ from fresh snapshots of different ages. Therefore the $\nu(t) \rightarrow 1$ observed here is a **threshold artifact**, *not* a confirmation of the claim about ν^{theor} : by § S1.6 / § S3.2, as soon as the fresh influx saturates the tracking floor, $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$, not to 1. The series illustrates precisely the mechanism $\eta_v \rightarrow 0$ / the inflation of I_{mem} (and the identity $\phi \sim \alpha\tau_E/t$), but does *not test* the ν^{theor} claim: its test requires a real joint Kalman estimate over the fresh FIFO subset, showing a ν^{theor} plateau at the level $\nu_{C1} < 1$ for $\alpha > 1$ — this was carried out by the separate non-circular series `superlinear_noncirc` (§ S4.4a). We do not remove the `superlinear_memory` series — we merely correctly delimit what it shows (the mechanism $\eta_v \rightarrow 0$, not the ν^{theor} plateau). *Window-convergence diagnostics* (`gamma_window_convergence` in `results_summary`; deterministic tabulation of the exact $\phi = 1 - ((t - \tau_E)/t)^\alpha$, not Monte Carlo): as the fit window (the decade $[w, 10w]$) slides to the right, the exponent γ grows monotonically toward 1 for *every* α — with the window raised by an order of magnitude ($w = 2 \cdot 10^4$) $\gamma \rightarrow \{1.000; 0.999; 0.998; 0.996\}$, at $\times 1000$ ($w = 2 \cdot 10^6$) $\gamma \rightarrow \{1.000; 1.000; 1.000; 1.000\}$ for $\alpha \in \{1; 1.5; 2; 3\}$; the local log-log slope of ϕ settles to -1 ($-0.897 \rightarrow -0.990 \rightarrow -0.999 \rightarrow \dots \rightarrow -1.000$ at $t = 2 \cdot 10^3 \dots 2 \cdot 10^9$). Thus the downward drift of γ on the baseline window is a subleading curvature of the finite window, not a dependence of the leading exponent on α ; the leading exponent is identically 1, and there is no polynomial α_c . For exponential growth (at $\kappa\tau_E \in \{0.1; 0.5; 1\}$) — $\phi_\infty = 1 - e^{-\kappa\tau_E} \in \{0.095; 0.394; 0.632\} > 0$, i.e. $\nu^{\text{theor}}(t) < 1$ (escape from the nostalgia floor), but the efficiency $\eta_v(t) = I_{\text{pred}}/I_{\text{mem}} \rightarrow 0$ *structurally*: the horizon-limited numerator $I_{\text{pred}} \leq H_\tau$ is divided by the unboundedly growing denominator I_{mem} (truly $\leq H(X_E^{\leq t})$, $\sim$ linearly, § S3.4; in the run taken conservatively $\propto |M_{\leq t}| = M_0 e^{\kappa t}$). The retained capacity grows to $|M_{\leq t}|(T)/|M_{\leq t}|(0) \in \{2.0 \cdot 10^4; 3.1 \cdot 10^{21}; 9.8 \cdot 10^{42}\}$ per run, and the *capacity-estimated* ratio of efficiencies collapses exactly inversely to it: $\eta_v(T)/\eta_v(0) \in \{5.0 \cdot 10^{-5}; 3.2 \cdot 10^{-22}; 1.0 \cdot 10^{-43}\} =$ the inverse capacity ratio $|M_{\leq 0}|/|M_{\leq T}|$ (S3.5–S3.6; in the run I_{mem} is taken $\propto |M|$, $I_{\text{pred}} = H_\tau$ — a *conservative upper estimate* of the decline; the true $I_{\text{mem}} \leq H(X_E^{\leq t})$ grows $\sim$ linearly, the actual decline is $\eta_v \sim 1/t$, the limit $\eta_v \rightarrow 0$ being the same) — the structural smallness of η_v , without appeal to the energy budget. The saturation $\phi(t) \rightarrow 0$ under polynomial growth (the absence of α_c) is shown in Fig. 4 of the main text.

S4.4a superlinear_noncirc (Theorem 3, non-circular test of the ν^{theor} -plateau at the C1 floor)

The series fixes the circularity of the threshold model of § S4.4/§ S4.5: instead of an age gate “a bit is useful $\Leftrightarrow \text{age} < \tau_E$ ”, the learner builds a *real joint Gaussian (Kalman/RLS) posterior* over the *current* θ_t from the growing fresh FIFO subset, so that several fresh snapshots of one coordinate *jointly* concentrate the estimate toward the tracking floor Σ_∞ — there is no inter-age gate. The model is the A1 linear-Gaussian representative of the softmax chain (§ S3.2): $K = 56$ independent coordinates $X_{t+1} = \theta_t X_t + w$

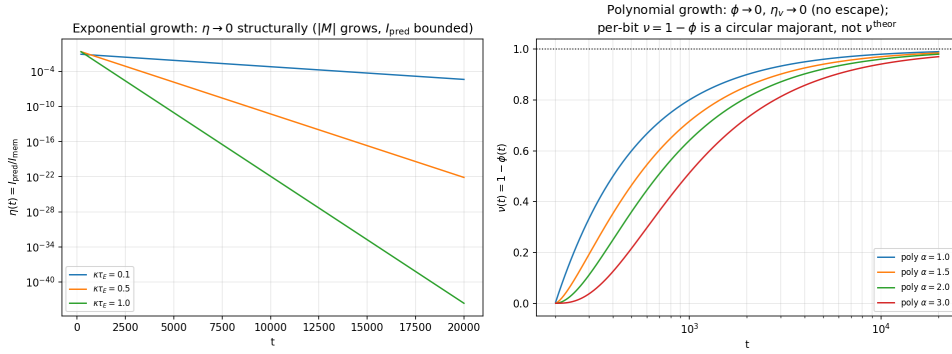


Figure S5. Series `superlinear_memory`, exponential escape and the structural smallness of η_v . With $|M_{\leq t}| = M_0 e^{\kappa t}$ the fresh fraction reaches a nonzero limit $\varphi_\infty = 1 - e^{-\kappa\tau_E} \in \{0.095; 0.394; 0.632\}$ (escape: $\nu^{\text{theor}} < 1$), but the retained capacity inflates to $|M_{\leq t}|(T)/|M_{\leq t}|(0)$ up to $9.8 \cdot 10^{42}$; the denominator I_{mem} grows unboundedly ($\leq H(X_E^{\leq t})$, $\sim$ linearly; in the run taken $\propto |M_{\leq t}|$ as a conservative estimate) while $I_{\text{pred}} \leq H_\tau$ is horizon-limited, so that $\eta_v \rightarrow 0$ structurally (the capacity-estimated $\eta_v(T)/\eta_v(0)$ down to $1.0 \cdot 10^{-43}$; the true decline is $\sim 1/t$). The escape formally keeps ν^{theor} below 1, but informationally — memory $\gg$ prediction, and η_v is negligible. The collapse of η_v is structural (growth of the denominator), not an energy-budget divergence.

($w \sim N(0, q)$, $q = 1$) with OU drift of the coupling $\theta_{t+1} = (1 - \lambda)\theta_t + u$ ($u \sim N(0, \sigma^2)$); θ is a transition coupling observed *only multiplicatively* through the pair (X_t, X_{t+1}) (the per-step Fisher information about θ_t is X_t^2/q , so the Fisher-rate $\mathcal{I}_{\text{rate}} = \mathbb{E}[X^2]/q$ is *measured*, not postulated). $\lambda = 0.1$, $\sigma^2 = 2 \cdot 10^{-3}$, $\sigma_0^2 = \sigma^2/(2\lambda) = 0.010$ (discretely-exact 0.0105), $T = 10^5$, $M_0 = 100$, $\alpha \in \{1; 1.5; 2; 3\}$, $\tau_E = 1/\lambda = 10$, 48 realizations, seed 20260527, numpy only; the plateau is averaged over $t \geq 10^4$.

Actual output (key result). The measured $\mathcal{I}_{\text{rate}} = 1.022$, whence $\epsilon_{\text{track}} = \mathcal{I}_{\text{rate}}\sigma^2/\lambda^2 = 0.204$ and the analytic floor $\nu_{C1} = (2/\epsilon_{\text{track}})(\sqrt{1 + \epsilon_{\text{track}}} - 1) = 0.954$ ($\approx 1 - \epsilon_{\text{track}}/4 = 0.949$; S3.3b). The primary (Fisher-cancelled) nostalgia measure $\nu_{\text{varratio}} = \mathbb{E}[(\theta - \hat{\theta})^2]/\mathbb{E}[\theta^2]$ — which is precisely Σ_∞/σ_0^2 to leading order (S1.3, Fisher cancels) — reaches a **plateau of 0.963** (+0.97% relative to the analytics 0.954; slope $d\nu/d\ln t = +1.2 \cdot 10^{-3}$ — flat), and it is **exactly α -independent**: for $\alpha \geq 1$ the capacity $M_0 t^\alpha \geq t$ retains the entire history, the joint posterior is one and the same for all α , the floor Σ_∞ is unbeatable. Two alternative (subleading Fisher-weighted) measures give 0.970 (the literal problem $\mathbb{E}[D_{KL}]/I_{\text{opt}}$) and 0.946 (fully X^2 -weighted) — both within $\pm 2\%$ of ν_{C1} , the spread reflecting a model-dependent $O(1)$ constant (§ S3.2). The tracking-error estimate reaches **the floor, not zero**: $\hat{\Sigma} \rightarrow \Sigma_\infty = 0.0101$ (realized MSE 0.0101) — memory obsolete beyond τ_{conc} does not reduce the tracking error. The sanity controls are exact: with no memory ($\hat{\theta} = 0$) $\nu = 1.000$, the oracle ($\hat{\theta} = \theta$) $\nu = 0.000$, a single snapshot $\nu = 0.992 \approx 1$. **The discrepancy (load-bearing evidence):** the naive per-bit majorant $\nu_{\text{naive}} = 1 - \phi \rightarrow 1$ (0.9997–0.9999 over α) — *the old circular result of superlinear_memory* — diverges from the real plateau $\nu^{\text{theor}} = 0.963$ (Fig. S6), whereas $\eta_v = I_{\text{pred}}/I_{\text{mem}} \rightarrow 0$ (the separation of ν^{theor} and η_v , § 5 item 4 main).

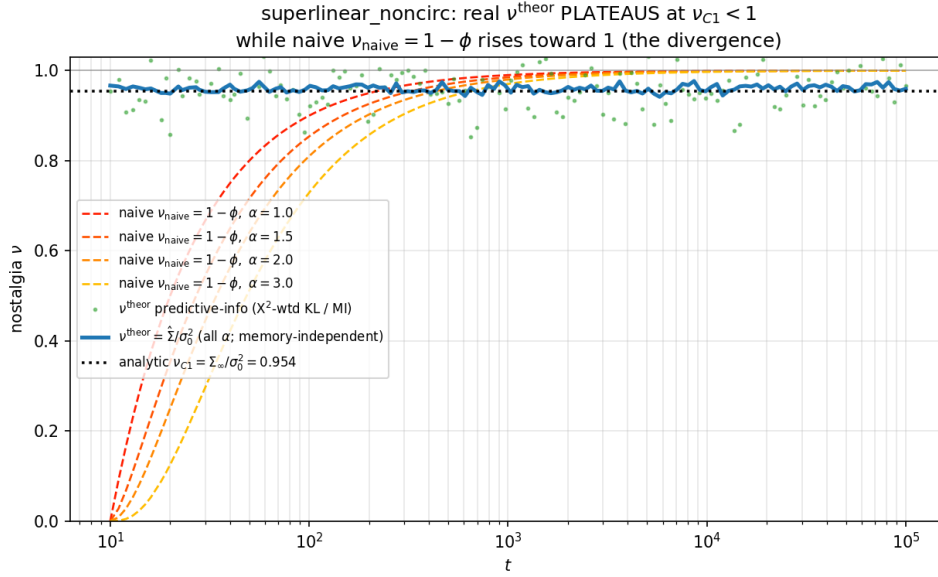


Figure S6. The `superlinear_noncirc` series: the ν^{theor} plateau at the C1 floor and its divergence from the naive majorant. The primary (Fisher-cancelled) $\nu^{\text{theor}} = \hat{\Sigma}/\sigma_0^2$ (solid) reaches a flat plateau at ≈ 0.96 , coinciding with the analytic $\nu_{C1} = \Sigma_\infty/\sigma_0^2 = 0.954$ (dotted), α -independently; the alternative predictive-info measure (dots) scatters around the same level. The naive per-bit majorant $\nu_{\text{naive}} = 1 - \phi$ (dashed, $\alpha \in \{1; 1.5; 2; 3\}$) instead rises toward 1 — the *circularity artifact* of the `superlinear_memory` threshold model (§ S4.4). The divergence of the solid from the dashed curves is the load-bearing evidence: the real ν^{theor} stands at $\nu_{C1} < 1$ rather than saturating toward 1.

All five validations (the plateau coincides with ν_{C1} ; $\hat{\Sigma} \rightarrow \Sigma_\infty \neq 0$; single snapshot $\rightarrow 1$; oracle $\rightarrow 0$; $\nu_{\text{naive}} \rightarrow 1$) — PASS. Thus the $\nu^{\text{theor}} \rightarrow 1$ of the § S4.4 threshold model is confirmed as a *circularity artifact*, while the real plateau stands at the C1 tracking floor $\nu_{C1} < 1$ (§ S1.6, § S3.2), α -independently. Results — `results_summary.{txt,json}`.

S4.5 General conventions and limitations

In all series the true $\theta_v(t)$ is known by construction, so $\nu^{\text{theor}} = \nu^{\text{op}}$ [1] § 6; $\nu(t)$ without a subscript is used. *Caveat on the estimator.* In the numerical series of #3 (`fou_floor`, `psp_floor`) nostalgia $\nu(t)$ is read not by a KSG mutual-information estimator but from the *threshold DPI horizon model*: a bit of age Δ is deemed useful if its predictive value by the realized autocorrelation $\hat{\rho}^2(\Delta)$ (survival on known $\theta_v(t)$) exceeds the DPI threshold $\rho^2(\tau_E)$, otherwise nostalgic. Since $\theta_v(t)$ is known by construction, conditioning on the latent is realized by *fixing* the true θ_v , and the conditional $I_{\text{pred}} = I(X_t; X_{t+\tau} \mid \theta_v(t))$ is measured directly — whence $\nu^{\text{theor}} = \nu^{\text{op}}$ by construction; a separate CMI estimator is not required. The mention of KSG [1] § S6 is inherited from the accompanying work #2 and does *not apply* to the numerical series of #3 (there θ_v is known by construction). *Circularity of the threshold model in `superlinear_memory` (important).*

The same threshold model (“a bit is useful $\Leftrightarrow$ age $< \tau_E$ ”) in the `superlinear_memory` series *hard-wires* per-bit uniformity and forbids joint estimation of $\theta_v(t)$ from fresh snapshots of different ages; therefore its $\nu(t) \rightarrow 1$ is a threshold artifact, not an independent test of the claim about ν^{theor} (which gives $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$, § S1.6, § S3.2). The series illustrates the mechanism $\eta_v \rightarrow 0$ / the inflation of I_{mem} , but does not test the ν^{theor} plateau at the level ν_{C1} — this is done by the separate non-circular series `superlinear_noncirc` (§ S4.4a), where a real joint Kalman posterior over the fresh FIFO gives a plateau $\nu^{\text{theor}} \rightarrow \nu_{C1} < 1$ coinciding with the analytic Σ_∞/σ_0^2 . The simulations illustrate but do not prove the theorems (§ S1–S3 carry the proofs); do not go beyond abstract Markov chains; do not cover the non-mixing (heavy-tailed) class outside the domain of Theorem 2. The full implementation, seeds, pseudocode, and figure regeneration are in the per-series `README.md` files (`simulations/*`) of the mirror repository `andriishin/landauer-undertow-oa`.

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